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A tutorial-based survey on feature selection: Recent advancements on feature selection

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ABSTRACT

Curse of dimensionality is known as big challenges in data mining, pattern recognition, computer vison and machine learning in recent years. Feature selection and feature extraction are two main approaches to circumvent this challenge. The main objective in feature selection is to remove the redundant features and preserve the relevant features in order to improve the learning algorithm performance. This survey provides a comprehensive overview of state-of-art feature selection techniques including mathematical formulas and fundamental algorithm to facilitate understanding. This survey encompasses different approaches of feature selection which can be categorized to five domains including: A) subspace learning which involves matrix factorization and matrix projection, B) sparse representation learning which includes compressed sensing and dictionary learning, C) information theory which covers multi-label neighborhood entropy, symmetrical uncertainty, Monte Carlo and Markov blanket, D) evolutionary computational algorithms including Genetic algorithm (GA), particle swarm optimization (PSO), Ant colony (AC) and Grey wolf optimization (GWO), and E) reinforcement learning techniques. This survey can be helpful for researchers to acquire deep understanding of feature selection techniques and choose a proper feature selection technique. Moreover, researcher can choose one of the A, B, C, D and E domains to become deep in this field for future study. A potential avenue for future research could involve exploring methods to reduce computational complexity while simultaneously maintaining performance efficiency. This would involve investigating ways to achieve a more efficient balance between computational resources and overall performance. For matrix-based techniques, the main limitation of these techniques lies in the need to tune the coefficients of the regularization terms, as this process can be challenging and time-consuming. For evolutionary computational techniques, getting stuck in local minimum and finding an appropriate objective function are two main limitations.

1. Introduction

This era can be called data explosion era with the rate of information generation is continuously increasing. That is why the term of big data has become quotidian term in information technology (IT), signal processing, machine learning, pattern recognition and digital health. Curse of dimensionality is become a well-known problem due to the increasing amount of information. Subsequently, dimensionality reduction and feature selection have been significantly gotten attention since the performance of machine learning algorithms can be considerably affected by the number of features and dimensions of data. Feature selection aim is to find the small subset of most discriminative information which carry the main information of high-dimensional data. This aim can be obtained by removing redundant features and maintaining the most relevant features. Uncorrelated features (multicollinearity) paly role as noise during training process, yielding in increased computational costs and potential degradation in the performance of the learning algorithm (Khalid, 2014; Ding, 2005). This fact is demonstrated in peaking phenomenon (Sima, 2008). Therefore, feature selection is categorized as one of the main preprocessing steps to generate a machine learning pipeline. Feature selection is significantly important for healthcare and medicine since the features are interpretable in medicine and health. Feature selection plays vital role for disease diagnosis (Makimoto et al., 2022), disease progression analysis (Moslemi et al., 2022b) and diseases discrimination (Moslemi et al., 2022a). For instance, Moslemi et al. showed that some lung quantitative computed tomography (qCT) features carry the more discriminative information to distinguish patients with chronic obstructive pulmonary disease (COPD) and patients with asthma. In another study, the feature selection is applied to classify patient with Covid-19 and those without

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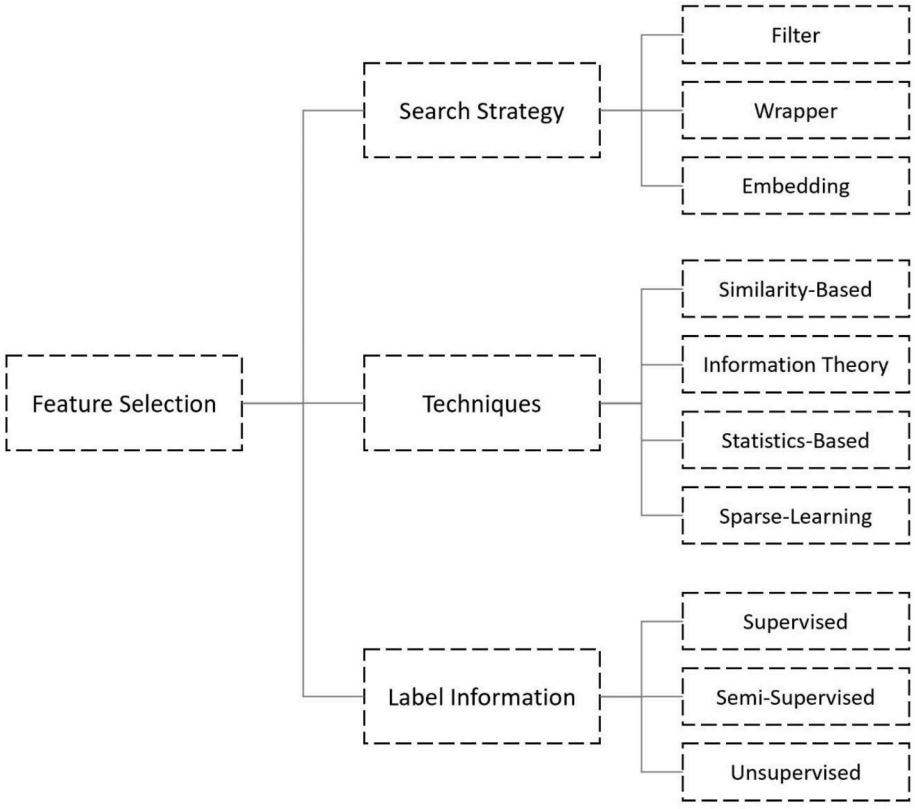


Fig. 1. Feature selection categories. The presented figure illustrates various types of feature selection and their subcategories.

(Saberi-Movahed et al., 2022b). In pharmacological studies, feature selection is employed to evaluate response to treatment (Mehrpooya et al., 2022). In oncology, feature selection is playing significant role for gene selection, and it improves the subsequent treatment (Ebrahimpour et al., 2018; Sun et al., 2019a,b; Sayed et al., 2019). In agriculture science, feature selection is utilized to find most discriminative information for different aims such as bacteria detection and water treatment (Buyrukoğlu, 2021a). In sport sciences, Feature selection can be used for positioning footballer (Buyrukoğlu and Savaş, 2023). These examples showed the importance of feature selection in different fields.

Feature selection techniques can be categorized based on strategy, labels information and techniques which is shown in Fig. 1.

Based on strategy, Feature selection techniques are taxonomized to Filter-based, Wrapper-based and embedded-based techniques.

1.1. Search strategy

1.1.1. Filter methods

In Filter methods, the features are ranked and assessed based on some criteria by statistical and intrinsic characters of the dataset. In filter method there is no connection between classifier and dataset. In this category, Features are ranked based on relevancy of features with target and irrelevant feature are filtered out based on correlation among the features (Kohavi, 1997; Guyon, 2003; P, 1994; Blum, 1997; John, 1994). Laplacian score (He, 2005), information theory-based techniques such as mutual information (Battiti, 1994; Lazar, 2012; Forman, 2003; Kwak and Choi, 2002), and maximum relevancy and minimum redundancy (Mrmr) (Peng et al., 2005a), statistical tests and matrix factorization-based technique (Elden, 2007; Halko et al., 2011) are common methods for feature selection.

1.1.2. Wrapper methods

In this strategy, the features are selected based on learning algorithm performance. Generally, the wrapper-based techniques are computationally expensive. If we consider N as the number of features for a dataset, there are 2^N subsets that can be candidate solution for this combinatorial problem. Therefore, the feature selection using wrapper methods is categorized as NP-hard problem. The optimum features are selected in iterative block such that the error of classifier is considered as criterion. Sequential forward feature selection, sequential backward feature selection, floating search (Pudil et al., 1994) and recursive support vector machine (SVM) (Zhang et al., 2006) are conventional wrapper-based feature selection techniques in machine learning. If learning algorithms encounter a large number of features, the main challenge of strategy-based methods lies in the exponential growth of computational time. To tackle this challenge, evolutionary algorithms can be applied for feature selection. Genetic algorithm (GA) (Leardi and Lupia'ñ ez Gonzalez, 1998; Leardi, 2000; Sikora and Piramuthu, 2007; Yang and Honavar, 1998; Frohlich and Chapelle, 2003), Ant colony (AC) (Kanan et al., 2007; Kanan and Faez, 2008; Ani, 2005; Hosseinzadeh Aghdam et al., 2009; Zhang and Hu, 2005), particle swarm optimization (PSO) (Chuang et al., 2008; Huang and Dun, 2008a), simulated annealing (SA) (Meiri and Zahavi, 2006), Grey wolf optimization (GWO) (Al-Tashi et al., 2019), gravitational search algorithm (GSA) (Rashedi et al., 2010) and Bat algorithm (BA) (Nakamura et al., 2012) are searching-based techniques which can be utilized to find optimum subset of features. These search algorithms initialize the selected features, and they move toward optimum solution based on classifier error. GA and SA are discrete algorithms and can be directly used for feature selection, whereas PSO, GWO, GSA and BA are continuous algorithm and must be binarized for feature selection goal or random key technique can be applied to deal this problem. In conclusion, since wrapper-based techniques are coupled to a classifier, its computational complexity is more expensive than filter-based technique.

1.1.3. Embedded methods

Incorporation of feature selection as part of training process can be considered as main difference of embedded strategy with filter and

wrapper strategies. Using greedy search in decision tree to spilite features based on information gain or Gini-index is an example of embedded strategy. Lasso regression (Tibshirani, 1996) is one of the well-known embedded. Group lasso feature selection as extended lasso technique was proposed to cope with lasso restrictions (Meier et al., 2008) and combination of group lasso with lasso was led to sparse group lasso technique (Simon et al., 2013). Elastic-Net is the combination of l_1 norm and l_2 norm regularizations that can be applied for feature selection (Zou and Hastie, 2005). Likewise, Linear discriminant analysis (LDA) is an embedded-based technique that aims to reduce the confusion between classes by maximizing the inter-class variance and minimizing the intra-class variance, thereby achieving dimensionality reduction (Fisher, 1936). Sparse regularization-based techniques have been broadly investigated, $l1$-SVM (Bradley and Mangasarian, 1998; Fung and Mangasarian, 2000) and l_1-logistic regression (Ng, 2004) are feature selection techniques which features are selected by l_1 sparse norm. In Hybrid Huberized SVM (HHSVM), l_1-norm and l_2-norm are combined to utilize the characteristics of l_1-norm and l_2-norm simultaneously (Wang et al., 2007). eXtreme Gradient Boosting (XGBoost) algorithm is a decision-based tree ensemble algorithm that is classified as embedded feature selection. In this context, re-weighting boosting feature selection based on fast and scalable tree-boosting models is the one of the recent studies in embedded strategy (Alsahaf et al., 2022).

1.2. Label information

Feature selection based on label information can be categorized to supervised methods (Wolf and Shashua, 2005), semi-supervised methods (Zhao and Liu, 2007), and unsupervised methods (Cai et al., 2010). Information about relevancy of features and target can be obtained in supervised based techniques using interaction between features and classes. Under semi-supervised condition, features are discriminated by extracting information from both labeled and unlabeled data, with a portion of the data being labeled. Most of the semi-supervised techniques leverage similarity matrix to select features by Laplacian graph (graph structure) construction (Huang and Dun, 2008a; Sheikhpour et al., 2018). There is no label information for unsupervised case and feature selection can be applied based on relevancy among features. Most of the time, this type of feature selection is regarded as subspace learning, as it aims to find a subspace of data that can span the original data (Dy and Brodley, 2004).

In addition to the aforementioned feature selection techniques, there are hybrid feature selection (Moslemi et al., 2022a) and ensemble feature selection (Buyrukoğlu, 2021b) methods as well. These techniques combine two or more feature selection techniques to obtain the most discriminative features. In the hybrid techniques, features are initially ranked by first technique, and then a top-k feature selection is performed. Subsequently, a second feature selection algorithm is applied to identify the most important features. For example, Moslemi et al. (2022b) introduced a hybrid feature selection technique that was combination of factor analysis and PSO. In the case of ensemble feature selection, multiple feature selection techniques are merged to identify the most significant features. BUYRUKOĞLU (Buyrukoğlu, 2021b) proposed an ensemble feature selection approach that combines Gain Ratio, ReliefF, Chi-Square, and fast correlation based filter (FCBF) methods to extract the most informative features. Ensemble feature selection can be modeled as homogeneous ensemble and heterogamous ensemble.

In this survey, feature selection techniques are classified to matrix-based techniques (subspace learning), sparse representation learning (compressed sensing and dictionary learning), information theory-based techniques, evolutionary computation techniques and reinforcement learning-based techniques.

In part 2, feature selection techniques based on matrix decomposition and regression problem are surveyed.

In part 3, sparse representation learning techniques for feature selection are discussed.

In part 4, information theory-based techniques for feature selection are discussed.

In part 5, evolutionary computation-based techniques such as GA, PSO, ACO and GWO for feature selection are discussed.

In part 6, reinforcement learning-based techniques for feature selection are discussed.

In part 7, A total of 22 benchmark datasets are presented, which can be utilized for evaluating various feature selection techniques.

The Taxonomy of feature selection techniques, which are surveyed in this study, is shown in Fig. 2.

2. Matrix-based techniques

Frequently used notations in this study are summarized in Table 1.

In feature selection using matrix-based techniques, we seek a subspace which has lower dimension compared to original space and it can span all information of original space. For a data-matrix $A := [f_1, f_2, \ldots, f_m]$, a new feature-set with J features $A_J := \{f_1, f_2, \ldots, f_J\}$, where $J < m$, can be found such that $span(A_J) = span(A)$. From linear algebra we know that if $span(A_J) = span(A)$, we will have $A_J \subset A$. Therefore, the objective function is to obtain a subspace which is sufficiently close to original space. It can be formulated as follows:

$$arg\min_J distance(span(A), span(A_J))$$

$$s.t \quad |J| = k \tag{1}$$

where k and J represent the number of selected features and index of selected features. Distance measurement between data and its subspace can be interpreted as feature selection. In the context of feature selection, there are three important criteria for subspace distance evaluation which are described as follows:

1) Hausdroff distance:

Hausdroff distance (HD) is proposed to evaluate distance between two sets (Shapiro and Blaschko, 2004).

$$\mathrm{HD}(U, V) = \max\{sup_{u\in U}\ inf_{v\in V}\|u - v\|, sup_{v\in V}\ inf_{u\in U}\|u - v\|\} \tag{2}$$

2) Directional distance:

Directional distance (DirDis) can be defined for two subspaces S_1 and S_2 as follows:

$$\mathrm{DirDis}(S_1, S_2) = \sqrt{\max(m, q) - Tr\left(V^T U U^T V\right)} \tag{3}$$

where $U \in R^{n\times m}$ and $V \in R^{p\times q}$ are standard basis matrices for S_1 and S_2, respectively (Wang et al., 2006).

Additionally, the angle between two vectors can be considered as a metric for similarity measurement. Therefor, the angle can be considered as a measure of the dissimilarity between two subspaces or features represented as vectors (Wolf and Shashua, 2003).

3) Frobenius Norm-base distance:

Selection a subset of features to represent all features can be considered as principle of feature selection. Then, we seek a subspace A_J which is a good approximate of original data A. In other words, feature selection can be defined as subspace of original data which is constructed by a small number of features. As a result, feature selection problem can be formulated as matrix-based optimization as follows:

$$\min_J \|A - A_J\|_F \quad s.t \quad |J| = k \tag{4}$$

Matrix factorization formula is not convex and then there exist no unique solution. There are some approaches to approximate matrix

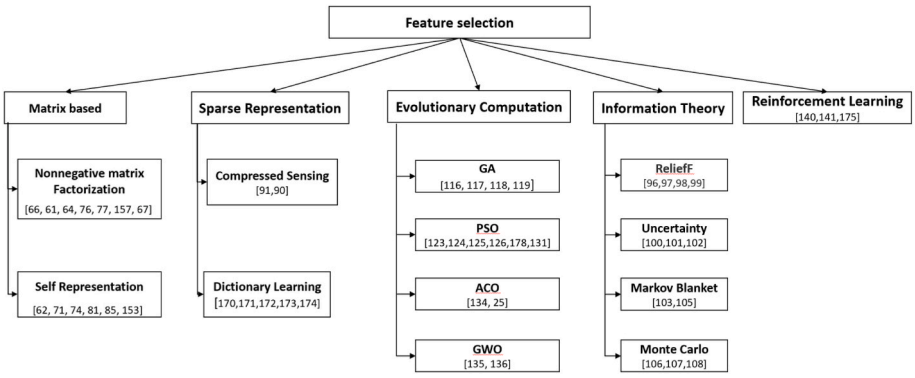


Fig. 2. The hierarchical structure of feature selection techniques based on the taxonomy of different feature selection approaches.

Table 1
Notations frequently used in this article.

Notation	Representation
n	Number of instances
m	Number of features
$\|J\|$	The cardinality of set J
J^c	the complement of set J
I	Unit matrix
u	Lower case shows the vector
A_{ij}	The (i,j)-th entry of matrix
A^T	The transpose of matrix A
$Diag(A)$	The vector of all diagonal elements of matrix A
$Tr(A)$	The trace of square matrix $Tr(A) = \sum_{i=1}^{n} A_{ii}$
$\|A\|_F$	Frobenius norm of A. $\|A\|_F = \sqrt{Tr(A^T A)}$
A	Matrix $A \in R^{n\times m}$
$\|u\|_p$	Vector norm. $\|u\|_p = (\sum_{i=1}^{n} u_i^p)^{1/p}$
$\|A\|_{q,p}$	Mixed matrix Norm. $\|A\|_{q,p} = (\sum_{i=1}^{m} \|a^i\|_q^p)^{1/p}$
$\mathbf{1}^{d\times d}$	Matrix with all elements equal to 1.
$\mathbf{1}_d$	Vector with all elements equal to 1.

factorization as convex problem by proposing some constraints. These techniques are Vector quantization (VQ) (Gersho and Gray, 2012), singular value decomposition (SVD) (Duda et al., 2012), principal component analysis (PCA) (Jolliffe, 2011), independent component analysis (ICA) (Spurek et al., 2018), concept factorization (CF) (Peng et al., 2018; Yan et al., 2017), nonnegative matrix factorization (NMF) (Lee and Seung, 1999, 2001).

In 2015, feature selection based on nonnegative matrix factorization (NMF) was proposed (Wang et al., 2015a). The concept behind Nonnegative Matrix Factorization (NMF) is based on the idea that a matrix can be approximated by the multiplication of two nonnegative matrices (Wang and Zhang, 2013). A matrix $A \in R^{n\times m}$ can be approximated by $W \in R^{n\times L}$ and $H \in R^{L\times m}$ as follows:

$$\min \|A - WH\|_F \;\; S.t \;\; W \geq 0, H \geq 0 \tag{5}$$

where W is a non-negative $n \times L$ feature weight matrix and H is a $L\times m$ representation matrix. The subspace matrix A_J in formula (4) can be represented by AW and problem (5) can be converted into feature selection problem by considering orthogonality constraint for matrix W (Wang et al., 2015a). Consequently, feature selection based on NMF is formulated as follows:

$$\min \|A - AHW\|_F \;\; S.t \;\; W \geq 0, H \geq 0, W^T W = I \tag{6}$$

where $W^T W = I$ is applied to hold matrix W as an indicator matrix. W as indicator matrix is defined as follows:

$$W_{i,j} = \begin{matrix} 1, & the\ j-th\ element\ of\ J\ is\ i, \\ 0, & otherwise \end{matrix} \tag{7}$$

Then, by adding orthogonality constraint to objective function as a regularization term we have:

$$\min_{W,H} \frac{1}{2}\|A - AHW\|_F^2 + \frac{\alpha}{4}\left\|W^T W - I_k\right\|_F^2 \quad s.t \quad W \geq 0, H \geq 0 \tag{8}$$

Parameter $\alpha > 0$ is a penalty term to make a trade-off between error of reconstruction and to keep the indicator matrix W close to orthogonality. Different α were used to get highest accuracy and results showed that the higher value of α guarantee the more orthogonality. Solving (8) is difficult and the Lagrange multiplier technique is utilized to relax optimization problem (8). Lagrange multipliers β and λ are considered for W and H, respectively. Therefore, Lagrange function is constructed as follows:

$$\mathscr{L}(W,H,\beta,\lambda) = \frac{1}{2}\|A - AHW\|_F^2 + \frac{\alpha}{4}\left\|W^T W - I_k\right\|_F^2 + Tr(\beta W) + Tr\left(\lambda H^T\right) \tag{9}$$

and (9) can be expanded as follows,

$$\begin{aligned} \mathscr{L}(W,H,\beta,\lambda) &= \frac{1}{2}Tr\left((A - AHW)^T (A - AHW)\right) \\ &+ \frac{\alpha}{4}Tr\left(\left(W^T W - I_k\right)^T \left(W^T W - I_k\right)\right) + Tr(\beta W) + Tr\left(\lambda H^T\right) \end{aligned} \tag{10}$$

Alternating process is applied to solve (10). In this technique, derivative with respect to a variable is applied on (10) such that rest of the variables considered fix.

By taking derivative of $\mathscr{L}(W,H,\beta,\lambda)$ with respect to W,

$$\frac{\partial \mathscr{L}}{\partial W} = \frac{1}{2}\left(-2A^T AH^T + 2A^T AWHH^T\right) + \alpha\left(WW^T W - W\right) + \beta = 0 \tag{11}$$

$$A^T AWHH^T + \alpha WW^T W + \beta = A^T AH^T + \alpha W \tag{12}$$

and with respect to H.

$$\frac{\partial \mathscr{L}}{\partial H} = \frac{1}{2}\left(-2W^T A^T A + 2W^T A^T AWH\right) + \lambda = 0 \tag{13}$$

$$W^T A^T AWH + \lambda = W^T A^T A \tag{14}$$

According to K.K.T conditions, $\beta_{ij} W_{ij} = 0$ and $\lambda_{ij} H_{ij} = 0$ for all $i \in [n]$ and $j \in [m]$. Consequently, W and H are updating in each iteration as follows:

$$W_{ij} \leftarrow W_{ij} \frac{\left(A^T AH^T + \alpha W\right)_{ij}}{\left(A^T AWHH^T + \alpha WW^T W\right)_{ij}} \tag{15}$$

$$H_{ij} \leftarrow H_{ij} \frac{\left(W^T A^T A\right)_{ij}}{\left(W^T A^T AWH\right)_{ij}} \tag{16}$$

The Algorithm of feature selection using NMF is summarized in Algorithm 1.

Algorithm 1. NMF feature selection.

Input: Data matrix $A \in R^{n\times m}$, parameter $\alpha > 0$ and k .

Output: Top k Features

Filter phase-NMF

1- Initialize W and H as identity matrix, and calculate $J = rank(A)$;

2- Repeat

3- Normalized $W = WD^{-1}$ where $D = diag(W^TW)^{\frac{1}{2}}$

4- Fix H and update W by $W_{ij} \leftarrow W_{ij}\frac{(A^TAH^T+\alpha W)_{ij}}{(A^TAWHH^T+\alpha WW^TW)_{ij}}$

5- Fix W and update H by $H_{ij} \leftarrow H_{ij}\frac{(W^TA^TA)_{ij}}{(W^TA^TAWH)_{ij}}$

6- Until Convergence

Calculate all Euclidian norm for each row of W and sort them in descending order to peak J top features.$A^J \in R^{n\times J}$,

After convergence, rows of feature weight matrix W are sorted based on Euclidean norm in descending order ($\|w^i\|_2$). i indicates the feature index in matrix A. Rank $\|w^{(1)}\|_2$, $\|w^{(2)}\|_2$, …, $\|w^{(m)}\|_2$ in descending order and arrange features of A corresponding to order of $\|w^{(i)}\|_2$ values. Then select top r-feature from the sorted dataset. The complexity of this method is $O(m^2)$. As aforementioned, NMF can be applied for feature selection if feature weight matrix W is indicator matrix. However, Farid et al. (Saberi-Movahed et al., 2019) using $W^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix}$ example showed that orthogonality condition ($W^TW = I_k$) is not sufficient to have indicator matrix.

To improve the performance of NMF feature selection, maximum projection and minimum redundancy feature (MPMR) selection was proposed (Wang et al., 2015b). This technique removes redundant features while keeping relevant features. The maximum projection term is obtained by $\|A - AHW\|_F^2$ such that HW is the orthogonal projection matrix to project A to subspace of J. The minimum redundancy is defined by redundancy rate of a feature subset. Redundancy of features can be measured using Pearson correlation between features as follows:

$$RED(J) = \frac{1}{k^2}\sum_{p,q\in J}\rho(p,q) \tag{17}$$

Therefore, MPMR cost function can be constructed by merging maximum projection term and redundancy term.

$$\|A - AHW\|_F^2 + \alpha RED(J) \tag{18}$$

Since the correlation can be defined as the normalized covariance of the feature matrix, the minimum redundancy term can be expressed as follows:

$$RED(J) = \frac{1}{k^2}Tr\left(W^TA^TAW\mathbf{1}_{k\times k}\right) \tag{19}$$

where $\mathbf{1} \in R^{k\times k}$ is a matrix, whose elements are ones and $|J| = k$.

Eventually, the cost function of MPMR feature selection can be expressed as follows:

$$\min_{W,H}\frac{1}{2}\|A-AHW\|_F^2 + \beta Tr\left(W^TA^TAW\mathbf{1}\right) + \frac{\alpha}{4}\|W^TW - I_k\|_F^2 \; s.t\; W\geq 0, H\geq 0 \tag{20}$$

$RED(J)$ enhances the ability to remove redundancy of NMF for feature selection. Similar to (11), by applying derivative with respect to W with fixed H, we have:

$$W_{ij}\leftarrow W_{ij}\frac{\left(A^TAH^T + \alpha W\right)_{ij}}{\left(A^TAWHH^T + 2\beta A^TAW\mathbf{1} + \alpha WW^TW\right)_{ij}} \tag{21}$$

and, derivative with respect to H with fixed W is led to (16). Same as NMF feature selection, the rows of W are ranked based on Euclidean norm and top r-feature from the sorted dataset. Total time complexity of MPMR is $O(m^2)$.

Regularization can improve the ability of feature selection. Generally, local learning regularization and global learning regularization are added to preserve local and global information.

In the context of local learning regularization, which is also called structure learning, geometrical information (Topology of data) of data can be preserved using local manifold embedding with aim of increasing the nonlinear ability of feature selection. Geometrical information of samples can be encoded by graph regularization. In this technique, connection among samples is encoded in a matrix which shows the connected samples. This regularization is applied to preserve information of samples after projecting into subspace. Loosely speaking, if two samples of A are close, corresponding samples in AW should be close (Cai et al., 2011; Belkin et al., 2001). Heat kernel function is frequently utilized to construct the affinity matrix S of samples. The affinity matrix for two samples x_i and x_j can be computed as follows:

$$S_{ij} = \exp\text{CHECKCHARACTERNOTSUPPORTED}\square\left(\frac{-\|a_i - a_j\|_2^2}{2\mathscr{E}^2}\right) \in R^{n\times n}$$

where i,j are the sample i and sample j, and $\mathscr{E}$ is the Gaussian scale parameter and it is recommended (Ren et al., 2014) to be set:

$$\mathscr{E} = \sum_{i,j}^{n}\frac{\sqrt{\|x_i - x_j\|_2^2}}{n^2}$$

Consequently, local geometrical information based on manifold learning can be derived as follows:

$$\sum_{i,j}\left(\|W^Ta_i - W^Ta_j\|_2^2 S_{ij} = \sum_{i=1}^{n}\left(W^Ta_i\right)^TW^Ta_iD_{ii} - \sum_{i,j=1}^{n}\left(W^Ta_i\right)^TW^Ta_jS_{ij}\right.$$

$$= Tr\left(W^TA^TDAW\right) - Tr\left(W^TA^TSAW\right) = Tr\left(W^TA^TLAW\right) \tag{22}$$

where $D = \sum_{j=1}^{n} S_{ij}$ is a diagonal matrix (degree matrix) and $L = D - S$ is the Laplacian matrix. Therefore, structural learning (local learning) regularization term is considered as follows:

$$\text{Local Learning Regularization} = Tr\left(W^T A^T L A W\right) \tag{23}$$

In the context of local learning, structure learning regularization term is added to cost function of NMF feature selection to preserve data geometrical information (Wang et al., 2020). Neighborhood preservation is another type of structure learning regularization which is added to NMF cost function (Wang et al., 2020). Neighborhood preservation is based on the assumption that any sample can be represented by a linear combination of its k-nearest neighbors. As a consequence, the objective function of NMF feature selection with structure learning can be expressed as follows:

$$\min_{W,H} \frac{1}{2}\|A - AHW\|_F^2 + \frac{\alpha}{4}\left\|W^T W - I_k\right\|_F^2 + \frac{\beta}{2} Tr\left(W^T A^T S A W\right)$$

$$s.t \quad W \geq 0, H \geq 0 \tag{24}$$

where S is a structure learning regularizer which is used for local learning goal, α is a coefficient to keep a trade-off between reconstruction error and local structure preserving and β is a coefficient to control the orthogonality degree of W. The affinity matrix of data is calculated by neighborhood preservation. The affinity matrix Z with Boolean entries can be computed as follows:

$$Z_{i,j} = \begin{cases} 1, if\ a_j \in N_k(a_i)\ \ or \quad a_i \in N_k(a_j) \\ 0, otherwise. \end{cases} \tag{25}$$

where $N_k(a)$ is the set of the k-nearest neighbors of sample a. Therefore, S neighborhood preservation can be derived by affinity matrix Z computation ($S = (I - Z)^T (I - Z)$).

By converting (24) to Lagrange function and applying derivative with respect to W, the update rule of W is

$$W_{ij} \leftarrow W_{ij} \frac{\left(A^T A H^T + \alpha W\right)_{ij}}{\left(A^T A W H H^T + \beta A^T S W A + \alpha W W^T W\right)_{ij}} \tag{26}$$

and H is same as (16).

Additionally, global learning regularization plays significant role to preserve feature weight matrix W as indicator matrix. Usually, sparse norm regularization terms such as mixed matrix norm $\|W\|_{q,p}$ and inner product norm are employed to achieve sufficient sparsity for W.

In another study, local learning and global learning regularization terms are added (Wang and Wang, 2017). This technique utilized self representation cost function and it is defined as follows:

$$\min_{p} \frac{1}{2}\|A - AP\|_F^2 + \alpha F(P) \tag{27}$$

where $P \in R^{m \times m}$ represents the coefficient matrix to assign selected features, $F(P)$ is the regularization function, $\|A - AP\|$ is self representation cost function and α is a coefficient to keep a trade-off between error of projection and regularization term. By adding structure learning and sparse $\ell_{2,1}$ norm terms to objective function (27), the unsupervised feature selection problem can be reformulated as follows:

$$\min_{p} \frac{1}{2}\|A - AP\|_F^2 + \frac{\beta}{4} Tr\left(P^T A^T S A P\right) + \alpha \|P\|_{2,1}$$

$$s.t \ \ rank(P) = k \tag{28}$$

where S is a structure learning regularization term, β is a coefficient to adjust the weight of structure learning, α is a regularization parameter to control sparsity. S is determined by graph Laplacian technique such that $S = (I - Q)^T (I - Q)$. Where Q is a symmetric affinity matrix of A and can be computed in several approaches such as Boolean value, Cosine function and heat kernel function (22). $\|P\|_{2,1}$ is applied to preserve sparsity and robustness against outliers. Rank constraint is considered for this feature selection technique to restrict selected features to linearly independent features. $rank(P) = rank(PP^T)$ and there is an assumption that the rank of P is equal to the number of selected features which shows the independent features. Consequently, there are $m - k$ eigenvalues equal to zero which represents the null space of matrix A. Therefore, it can be inferred that the sum of these m-k eigenvalues is zero. Based on Ky Fan's Theorem, the sum of zero eigenvalues ($\sum_{i=1}^{m-k} \lambda(PP^T) = 0$) can be approximated as follows (Fan, 1949):

$$\sum_{i=1}^{m-k} \lambda\left(PP^T\right) = \min Tr\left(G^T PP^T G\right) \ \ s.t\ G^T G = I \tag{29}$$

where $G \in R^{m \times (d-k)}$ and λ is eigenvalue of PP^T.

The final cost function can be written as follows:

$$\min_{p} \frac{1}{2}\|A - AP\|_F^2 + \frac{\beta}{4} Tr\left(P^T A^T S A P\right) + \alpha \|P\|_{2,1} + \frac{\xi}{2} Tr\left(G^T PP^T G\right)$$

$$s.t\ G \in R^{m \times (d-k)}, G^T G = I \tag{30}$$

The cost function (30) can be solved using iterative algorithm for P and in each iteration G is updated by eigenvectors corresponding to $m - k$ eigenvalues of PP^T.

Optimization problem of traditional least squares regression is based on ℓ_2 or Frobenius norm. $\|.\|_2^2$ and $\|.\|_F^2$ suffer from outliers and noise reinforcing. Aiming to tackle these defects, the $\ell_{2,1}$ norm is proposed to suppress the noise and outlier effects. This matrix norm has received significant attention for feature selection task due to joint sparsity ability of finding solution. Regarding the applications of $\ell_{2,1}$ norm in feature selection, group lasso and logistic group lasso were proposed for feature selection. $\ell_{2,p}$, where $0 < p < 1$, increases sparsity of solution, but *it* is non-convex and Lipschitz discontinuous function and these are big challenges for solving via gradient technique. To circumvent this challenge, a unified algorithm $\min_{W} loss(W) + \lambda F(W)$ was proposed (Wang et al., 2014). Where $loss(W), F(W)$ and λ ($\lambda > 0$) are loss function, regularization function and regularization parameter, respectively. $loss(W)$ and $F(W)$ are determined based on data distribution and application (Nie et al., 2010). For feature selection application the objective function is expressed follows:

$$\min_{X} \|AX - Y\|_{2,p} + \lambda F(X) \tag{31}$$

where $X \in R^{m \times c}$, $A \in R^{n \times m}$ and $Y \in R^{n \times c}$ are, projection matrix, data matrix and label matrix, respectively. Where n, m and c are number of samples, number of feature and number of classes, respectively. $F(X)$ is regularization term which can be sparse $\ell_{2,p}$ norm. Then, a robust feature selection based on matrix norm $\ell_{2,1}$ was proposed (Nie et al., 2010). The cost function based on $\ell_{2,1}$ norm is expressed as follows:

$$\min_{X} \|AX - Y\|_{2,1} + \lambda \|X\|_{2,1} \tag{32}$$

If a new term $AX - Y = \lambda E$ is defined, the (32) can be rewritten as follows:

$$\min_{X,E} \|E\|_{2,1} + \lambda \|X\|_{2,1} \quad s.t\ \ AX - Y = \lambda E \tag{33}$$

$$\min_{X,E} \left\| \begin{bmatrix} X \\ E \end{bmatrix} \right\|_{2,1} s.t \quad [A \quad \lambda I] \begin{bmatrix} X \\ E \end{bmatrix} = Y \tag{34}$$

$U = \begin{bmatrix} X \\ E \end{bmatrix} \in R^{(n+m) \times c}$ and $B = [A \quad \lambda I] \in R^{n \times (m+c)}$ as new terms for formula (34) were considered.

Finally, the objective function can be expressed as follows:

$$\min_{U} \|U\|_{2,1} s.t \quad BU = Y \tag{35}$$

Since solving the optimization problem (35) is computationally expensive and difficult, Lagrangian function technique can be utilized to drive a solvable mini-max optimization problem. The mini-max optimization problem by considering $\|U\|_{2,1} = Tr(U^TQU)$ can be expressed as follows:

$$\mathscr{L}(U,Q) = Tr\left(U^TQU\right) - Tr(\Lambda(BU-Y)) \tag{36}$$

$$Q = diag\left\{\frac{1}{2\|u^1+\varepsilon\|_2}, \frac{1}{2\|u^2+\varepsilon\|_2}, \cdots, \frac{1}{2\|u^{n+m}+\varepsilon\|_2}\right\} \tag{37}$$

where Q is a diagonal matrix ($Q \in R^{(n+m)\times(n+m)}$) and Λ is lagrange multiplier matrix. Optimization problem (36) can be solved for U using an iterative algorithm, and since (36) is convex the solution of U is a global optimum. In general, $\|U\|_{2,p} = Tr(U^TQU)$, $Q = diag\left\{\frac{p}{2\|u^1+\varepsilon\|_2^{2-p}}, \frac{p}{2\|u^2+\varepsilon\|_2^{2-p}}, \cdots, \frac{p}{2\|u^{n+m}+\varepsilon\|_2^{2-p}}\right\}$ and ε is very small value to avoid the occurrence of zero in the denominator (Wang et al., 2014).

In order to increase the theoretical justification of $\ell_{2,1}$ for feature selection aims, Rescaled Linear Square Regression (RLSR) was proposed for semi supervised feature selection (Chen et al., 2017). In this method, a set of scale factors are utilized to evaluate the importance of features. $Y \in R^{n\times c}$ is label matrix, since the problem is considered as semi-supervised, Y is partitioned to Y_l and Y_u for labeled data and missed labeled data, respectively. The objective function is constructed by similarity measurement (Frobenius norm) and $\ell_{2,1}$ norm regularization as follows:

$$\min_{X,\theta,b} \left\|A^T\theta X + \mathbf{1}b^T - Y\right\|_F + \lambda\|X\|_{2,1}$$

$$s.t$$

$$X, b, \theta > 0, \mathbf{1}^T\theta = 1, Y_U > 0, Y_U\mathbf{1} = \mathbf{1} \tag{38}$$

The objective function (38) is solved for fixed X and b to find $\theta = \frac{\|x_i\|}{\sum_{i=1}^{m}\|x_i\|}$. Therefore, Lagrangian function for each variable X, b and Y_u are expressed as follows:

$$\begin{cases} \mathscr{L}(b) = Tr\left(\left(A^TX + \mathbf{1}b^T - Y\right)^T\left(A^TX + \mathbf{1}b^T - Y\right)\right) \\ \mathscr{L}(X) = Tr\left(\left(A^TX + \mathbf{1}b^T - Y\right)^T\left(A^TX + \mathbf{1}b^T - Y\right)\right) + \lambda Tr\left(X^TQX\right) \\ \mathscr{L}(Y_u) = \frac{1}{2}\left(X^TA + b - Y\right)^T\left(X^TA + b - Y\right) + \delta\left(Y^T\mathbf{1} - 1\right) - Y^T\beta \end{cases} \tag{39}$$

where Q can be calculated based on (37) and δ, β are largange multipliers. X, b and Y are placed in iterative algorithm, and θ is calculated after convergence. θ shows the importance of features and it is sorted in descending order to select k top features.

To increase the sparsity of solution, Sparse Rescaled Linear Square Regression (RLSR) was proposed by wherein $\ell_{2,p}$ is substituted instead of $\ell_{2,1}$, where $p \in (0,1)$ (Chen et al., 2018). In this method, $\theta = \left(\frac{\|x_i\|}{\sum_{i=1}^{m}\|x_i\|}\right)^{\frac{1}{p}-0.5}$ and b is approximated like RLSR by (39). The variable p can be adjusted as hyperparameter to improve the performance. Finally, objective function can be expressed as follows:

$$\mathscr{L}(X) = Tr\left(\left(A^TX + \mathbf{1}b^T - Y\right)^T\left(A^TX + \mathbf{1}b^T - Y\right)\right) + Tr\left(\left(\theta^{-1}X\right)^T\left(\theta^{-1}X\right)\right) \tag{40}$$

The sparsity regularization term is the main difference between Sparse RLSR and RLSR. In RLSR the sparsity is adjustable to obtain the best performance.

However, it is important to highlight that the process of collecting labels can be expensive. Int this vein, regression model can be modeled as unsupervised model using laten representation learning. In latent representation learning, label matrix Y is approximated by adjacency matrix which is constructed based on samples distance. Tang et al. (2019) proposed latent representation learning and manifold regularization for unsupervised feature selection. Laplacian graph regularization and $l_{2,1}$ norm were employed to preserve topological information of samples and row-sparsity for feature selection, respectively. The latent representation learning and manifold regularization for unsupervised feature selection can be modeled as follows:

$$\min_{W,Y} \|XW - Y\|_F + \lambda\|W\|_{2,1} + \beta\left\|A - YY^T\right\|_F + \gamma Tr\left(W^TA^TSAW\right) \;\; s.t\; Y \geq 0 \tag{41}$$

where A and S are affinity matrix and Laplacian matrix, respectively. Y is Pseudo-label matrix and it can be calculated similar to affinity matrix S.

In another study, Shang et al. (2021) proposed dual space latent representation learning to consider information of feature space and data space simultaneously. This feature selection technique is modeled as following:

$$\min_{W,Y} \|XW - Y\|_F + \lambda\|W\|_{2,1} + \beta\left\|A - YY^T\right\|_F + \rho\left\|B - WY^T\right\|_F + \gamma Tr\left(W^TA^TSAW\right) \tag{42}$$

$$s.t\; Y \geq 0, W \geq 0, W^TW = I$$

where B and A are affinity matrices. Additionally, Wang et al. (2022b) embedded latent representation learning into autoencoder for unsupervised feature selection.

Graph Laplacian and neighborhood preservation are two main local structure learning techniques which are used to preserve geometrical information of data space after mapping to subspace. The most of techniques applied structure learning using fixed similarity matrix (Wang et al., 2020, 2022a; Wang and Wang, 2017; Zhu et al., 2023a; Zeng et al., 2022), which means that the affinity matrix is constructed based on input data (out of iteration body of algorithm) and never updated in iteration process. Consequently, the noisy data lead to noisy affinity matrix and subsequently feature selection performance is directly affected. To tackle this challenge, affinity matrix should be updated in each iteration. To this end, structured optimal graph feature selection (SOGFS) as an unsupervised feature selection was proposed in order to update affinity matrix in each iteration (Nie et al., 2016). This method consists of three main parts: local structure preserving, minimum redundancy consideration and sparsity. In this technique feature selection and structure learning are simultaneously performed with aim of local structure preserving. The connection among the samples is defined by a probability function to consider the local structure information. The similarity matrix S is defined such that each element of this matrix indicates similarity between two samples and it is symmetric matrix. The rank of data matrix must be minimized to have minimum redundancy.

Matrix $L_s = D - \frac{S^T+S}{2}$ is called graph Laplacian matrix and it was leveraged to preserve geometry information. D is diagonal matrix $D = diag\left(\sum_j \frac{s_{ij}+s_{ji}}{2}\right)$.Rank minimization problem is modeled using nuclear norm (sum of singular values) and it can be solved using Key Fan theorem as discussed in (29) and (30).

Finally, projection matrix W is defined as a mapping function (matrix) to project data in low-dimensional manifold and $l_{2,1}$ norm is added to cost function to have global structure learning regularization. Therefore, this technique can be mathematically expressed as follows:

$$\min \sum_{i,j}\left(\left\|W^Tx_i - W^Tx_j\right\|_2^2 s_{ij} + \alpha {s_{ij}}^2\right) + 2\lambda Tr\left(F^TL_sF\right) + + \gamma\|W\|_{2,1}$$

$$s.t$$

$$\|S\|_1 = \mathbf{1}, 0 \le s_{i,j} \le 1, F^T F = I, W^T W = I \tag{43}$$

where first term, second term and third term in (43) indicate local structure learning, minimum redundancy and sparsity, respectively. α, λ and γ are regularization terms. To solve (43), consider the F and S as constant variables and then solve for W and repeat for F such that W and S are fixed. This point should be noted that λ must be sufficiently large. Using alternating technique, (43) is solved for W as follows:

$$L(W,Z) = Tr\left(W^T A^T S A W\right) + \gamma \|W\|_{2,1} + Tr\left(Z(W^T W - I)\right) \tag{44}$$

where S and Z are similarity matrix and Lagrange multiplier, respectively.

To solve (44) for F, matrix W and S are considered fixed and (43) is converted to following:

$$\min_F Tr\left(F^T L_s F\right) \quad s.t. \quad F^T F = I \tag{45}$$

Lagrange function of (45) is

$$L(F,\psi) = Tr\left(F^T L_s F\right) - \psi\left(F^T F - I\right) \tag{46}$$

and setting $\frac{\partial L(F,\psi)}{\partial F} = 0$ we get:

$$L_s F = \psi \mathrm{F}. \tag{47}$$

Equation (47) indicates that F and ψ are eigenvector and eigenvalue of matrix L_s. Then, the optimal solution for variable F is equal to the c-eigenvector of L_s corresponding to the c smallest eigenvalues. Parameter c is a hyperparameter which shows the number of connected samples. In next step, problem (43) is solved for S to update affinity matrix in each iteration. To this end, matrices $M \in R^{n\times n}$ $(m_{ij} = \left\|W^T x_i - W^T x_j\right\|_2^2)$ and $N \in R^{n\times n}$ $(n_{ij} = \left\|f_i - f_j\right\|_2^2)$ and vector $d_{ij} = m_{ij} + \lambda_4 n_{ij} \in R^{n\times 1}$ are introduced to have (48) formula.

$$min \left\| s_i + \frac{1}{2\alpha} d_j \right\|_2^2 \quad s.t. \ \ s_i^T \mathbf{1} = 1, 0 \le s_{i,j} \le 1 \tag{48}$$

Using Lagrangian technique (51) can be expressed as follows:

$$L(s_i, \theta, \Phi_i) = \frac{1}{2}\left\| s_i + \frac{1}{2\alpha} d_j \right\|_2^2 - \theta\left(s_i^T \mathbf{1} - 1\right) - {\Phi_i}^T s_i \tag{49}$$

where θ and Φ_i are Lagrangian multipliers and the optimal solution of s_i is obtained using Karush-Kuhn-Tucker conditions. Lagrangian multiplier θ is

$$\theta = \frac{1}{k} + \frac{1}{2k\alpha_i} \sum_{j=1}^{k} d_{ij} \tag{50}$$

where k is the number of neighbors connected to i th sample and optimal solution of s_i is found as follows:

$$s_{ij} = \left(-\frac{1}{2\alpha_i} d_{ij} + \theta \right)_+ \tag{51}$$

consequently, the optimal α_i to have k neighbors connected to i th sample computed as follows:

$$\alpha_i = \frac{k}{2} d_{i,k+1} - \frac{1}{2} \sum\nolimits_{j=1}^{k} d_{ij} \tag{52}$$

By updating α_i in each iteration, each element of affinity matrix S is subsequently updated and updated Laplacian matrix L_s is obtained at end of each iteration. The most important point here is that if the input data is noisy or has outliers, the Laplacian matrix will be filtered in each iteration to consider high quality samples.

Adaptive Laplacian graph is embedded in NMF feature selection, and adaptive NMF feature selection was proposed (Shang et al., 2020a) which is called feature selection via adaptive structure learning and rank approximation. Formula (53) shows the objective of function of this technique.

$$\min \|A - AHW\|_F^{\ 2} + \gamma \|W\|_{2,1^{\frac{1}{2}}}^{\frac{1}{2}} + \sum_{i,j} \left(\left\|W^T x_i - W^T x_j\right\|_2^2 s_{ij} + \alpha {s_{ij}}^2 \right) + \lambda \sum_{i,j} \left\|f_i - f_j\right\|_2^2 s_{ij} \quad S.t$$

$$W \ge 0, H \ge 0, \forall i, \|S\|_1 = \mathbf{1}^d, 0 \le s_{i,j} \le 1, F^T F, F \in R^{n\times c} \tag{53}$$

The reason why $p = \frac{1}{2}$ was selected in this study is that experimental results indicate the lowest error of classification can be found by this value.

In another study, to mitigate the noise of similarity matrix, a probabilistic neighbors-based manifold structure preservation method was proposed to increase the preservation of local structure information and robustness against the noise. In this technique, a probability variable is defined as optimization problem, which is called z, to measure probability of connection between two samples, and similarity matrix Z is generated by solving optimization problem of z. Although obtaining the similarity matrix Z with rank constraint is difficult, graph Laplacian L enables to cope with this challenge. In the last step, a projection matrix W is defined to project original data matrix A into low-dimension manifold. The cost function can be expressed as follows:

$$\min_{W,H,G,Z} \|A - AHW\|_F^{\ 2} + \frac{1}{2} \sum_{i,j} \left(\left\|W^T a_i - W^T a_j\right\|_2^2 z_{ij} + \alpha z_{ij}^2 \right) + 2\gamma Tr\left(G^T L_z G\right) + \delta \|W\|_{2,\frac{1}{2}}^{\frac{1}{2}} + \frac{\mu}{2} \left\|W^T W - I\right\|_2^2 + Tr\left(\psi W^T\right) + + Tr\left(\varphi H^T\right)$$

$$s.t \forall i \ z_i^t 1 = 1, 0 \le z_{ij} \le 1, G^T G = I \tag{54}$$

where $\|W\|_{2,\frac{1}{2}^{\frac{1}{2}}} = Tr(W^T Q W)$ and $= diag\left(\frac{1/2}{2\|u^i + \varepsilon\|_2^{\ 3/2}}\right)$, and ψ , φ are lagarange multiplier diagonal matrices. a_i and a_j represent samples and parameter γ should be set sufficiently large because $Tr(G^T LG) = \sum_{ij} \left\|g_i - g_j\right\|_2^2 z_{ij}$ expected to be close to zero based on Ky Fans' theory. Additionally, γ should be adaptively adjusted such that if the number of features is less than assigned number, γ parameter should be increased and otherwise it should be decreased. α is a regularization parameter to guarantee no trivial solution, and μ and δ are balance parameters. Like NMF feature selection and $\ell_{2,p}$ feature selection techniques, W, H, G, Z can be optimized separately. In this technique, in each iteration, z, which is probability of connection between two samples, is updated. In order to update z, $r_{ij} = \left\|W^T a_i - W^T a_j\right\|_2^2$ and $t_{ij} = \left\|g_i - g_j\right\|_2^2$ are defined and then we have

$$\sum_{ij} \left(\left\|W^T a_i - W^T a_j\right\|_2^2 z_{ij} + \alpha z_{ij}^2 \right) + 2\gamma Tr\left(G^T L_z G\right) = r_{ij} z_{ij} + \alpha z_{ij}^2 + 2\gamma t_{ij} z_{ij}$$

and objective function for updating z can be formulated as follows:

$$min \left\| z_i - \frac{1}{2\alpha} h_i \right\|_2^2 \ s.t \ z_i^T \mathbf{1} = 1, 0 \le z_{ij} \le 1 \tag{55}$$

where $h_i = r_{ij} + \gamma t_{ij}$ and z_{ij} are vector with all elements one and probability of connection between two neighbors. This technique has many hyperparameters which must be tuned and obtaining optimum value for each of them can be a challenge.

Although $\ell_{2,1/2}$ regularization function obtains sparse solution, it is neither convex and nor Lipchitz continuous. Lipchitz discontinuity decreases control on gradient and it is considerable challenge for gradient-

based technique. To tackle this challenge, $\ell_{2,1-2}$ matrix norm was proposed that is nonconvex, but it is Lipchitz continuous (Shi et al., 2018). Nonconvexity challenge is still remained for this norm, but an iterative algorithm based on ConCave–Convex Procedure (CCCP) is designed to cope with this challenge. $\ell_{2,1-2}$ norm is defined based on difference of $\ell_{2,1}$ and $\ell_{2,2}$, and solution of this norm can be implied as natural improvement of $\ell_{2,1}$ sparse regularization. This technique is proposed for both supervised and unsupervised feature selection. In terms of supervised feature selection, the cost function can be expressed as follows:

$$\min_W L_{AY}(W) + \|W\|_{2,1-2} \tag{56}$$

where A, Y are data matrix and target, respectively. $L_{AY}(W)$ can be different loss functions such as Frobenius norm, logistic loss and multiclass hinge loss.

Since $\|W\|_{2,1-2} = \|W\|_{2,1} - \|W\|_{2,2}$ is not convex, linearization of $\|W\|_{2,2}$ via CCCP was proposed to combat this challenge. For matrix W, linearization is defined as follows:

$$\nabla\|W\|_{2,2} = \begin{cases} \left\{\dfrac{W}{\|W\|_{2,2}}\right\} & if\ W \neq 0 \\ 0, O.W & \end{cases} \tag{57}$$

And (56) can be rewritten as follows:

$$\min_W L_{AY}(W) + \alpha\|W\|_{2,1} - \alpha\|W\|_{2,2} \tag{58}$$

Using CCCP theory

$$\min_W \left(L_{AY}(W) + \alpha\|W\|_{2,1}\right) - \alpha < W, \nabla\|W\|_{2,2} > \tag{59}$$

$$\min_W \left(L_{AY}(W) + \alpha\|W\|_{2,1}\right) - \alpha Tr\left(W^T \|W\|_F^{-1} W\right) \tag{60}$$

For instance, the (56) can be written as follows by considering Frobenius norm as the loss function L_{AY} :

$$\min_W \left(\left\|Y - A^T W\right\|_F^2 + \alpha\|W\|_{2,1}\right) - \alpha Tr\left(W^T \|W\|_F^{-1} W\right) \tag{61}$$

Equation (61) can be solved like all the explained methods in this survey by getting derivative with respect to W and putting in an iteration algorithm. The ability of $\|W\|_{2,1-2}$ regularization to increase the performance of feature selection has been reported in different feature selection studies (Zhang et al., 2019b; Miao et al., 2021).

The use of sparse regularization initially considered the constraint of $\ell_{2,0}$, which is both NP-hard and nonconvex, making it difficult to solve. To overcome this challenge, $\ell_{2,1}$ regularization was introduced as an alternative. However, $\ell_{2,1}$ can be seen as a relaxed version of $\ell_{2,0}$. Subsequently, researchers proposed the regularization terms $\ell_{2,1/2}$ and $\ell_{2,1-2}$ to enhance the effectiveness of $\ell_{2,1}$. Nevertheless, these regularization terms select individual features one at a time instead of directly choosing a subset of features. To tackle this issue, Nie et al. (2020) presented a novel approach that allows for the use of the $\ell_{2,0}$ norm as a constraint for row sparsity. This is important to consider when dealing with features F_1, F_2, F_3 and F_4, where they are ranked as $F_1 > F_2 > F_3 > F_4$. While the top two features would be F_1 and F_2, it is possible that the combination of F_1 and F_3 in the subspace could be more discriminative. In (Nie et al., 2020), the problem (43) incorporated the $\ell_{2,0}$ constraint, and a novel solution was put forth to precisely identify the k most significant features. Another research study (Nie et al., 2022) introduced the integration of the $\ell_{2,0}$ constraint into linear discriminant analysis (LDA) as a method of supervised feature selection. Furthermore, Zhu et al. (2023b) combined the $\ell_{2,0}$ constraint with graph learning techniques for unsupervised feature selection.

Wang et al. (2014), Nie et al. (2010), Chen et al. (2017, 2018) and Shi et al. (2018) studies investigated about sparsity regularization term which is global structure learning regularization. Although sparsity regularization increases the sparsity of solution and decreases the effect of noise or outliers, local structure learning regularization term is required to reduce information losing due to dimensionality reduction as well. Therefore, global and local structure learning are playing significant role for feature selection aim simultaneously.

Although several studies proposed techniques to preserve the geometric information during feature selection, those methods only considered geometry information for data space. As a consequence, these methods can only preserve the local manifold information of the data space and feature space information cannot be preserved.

Meng et al. proposed dual-graph sparse non-negative matrix factorization (DSNMF) to tackle this challenge (Meng et al., 2018). Dual graph is considered to preserve geometric information of both the data space and the feature space. In this technique, the Laplacian matrix is calculated for both the data space and the feature space. The objective function DSNMF is mathematically formulated as follows:

$$\left\|A - WH^T\right\|_F^2 + \alpha\, Tr\left(W^T L^w W\right) + \beta\, Tr\left(H^T L^H H\right) + \theta\|W\|_{2,1}$$

$$s.t\ \ W \geq 0, H \geq 0 \tag{62}$$

where $\|W\|_{2,1}$ is added to increase sparsity on feature space and θ is a hyperparameter to balance between error of matrix factorization and noise. α and β are hyperparameters to balance between error of reconstruction and geometrical information preserving terms.

In order to preserve information in both data space and feature space, in other study, Farid et al. (Saberi-Movahed et al., 2022a) considered inn-product norm for both spaces and called it dual regularized unsupervised feature selection based on matrix factorization and minimum redundancy (DR–FS–MFMR). In contrast to DSNMF that considered structure learning for both spaces, DR–FS–MFMR considered sparse regularization for both spaces. They proposed dual regularized NMF feature selection based on inner product norm as sparse regularization for both data space and feature space and they reported the improvement of feature selection using dual regularized. As we discussed for $\|W\|_{2,1-2}$ and $\|W\|_{2,1/2}$, Convexity and Lipschitz discontinuity are the main challenges and $\ell_{2,1}$ norm is not appropriate regularization function when the high sparsity and low redundancy between rows are both necessary. That is why inner product regularization was introduced (Qi et al., 2018) which can preserve both indicator function constraint (using W rows correlation consideration) and sparsity of the weight matrix W. This norm is a convex norm and can be derived as follows:

$$\begin{aligned} inner-product\ Norm &= \sum_{i,j=1,i\neq j}^{d} < w^i, w^j > = \sum_{i,j=1,i\neq j}^{d} w^i w^{jT} \\ &= Tr\left(1_{d\times d} WW^T\right) - Tr\left(WW^T\right) \end{aligned} \tag{63}$$

Then inner-product norm is applied on feature weight matrix W and representation matrix H to preserve both spaces. The objective function of DR–FS–MFMR is expressed as follows:

$$\begin{aligned} &min\ \frac{1}{2}\|A - AHW\|_F^2 + \frac{\lambda_1}{2}\left(Tr\left(1_{d\times d} WW^T\right) - Tr\left(WW^T\right)\right) \\ &+\frac{\lambda_2}{k^2}\left(\mathbf{1}_{k\times K}^T W^T A^T AW \mathbf{1}_{k\times K}\right) + \frac{\lambda_3}{2}\left(Tr\left(1_{d\times d} H^T H\right) - Tr\left(H^T H\right)\right) \end{aligned}$$

$$s.t\ \ W \geq 0, H \geq 0 \tag{64}$$

where λ_1, λ_2 and λ_3 are coefficients and $corr = (\mathbf{1}_{k\times K}^T W^T A^T AW \mathbf{1}_{k\times K})$ is the correlation regularization to minimize the correlation among the features. In this study, direct correlation between representation matrix H and data A is shown. Based on NMF feature selection we have following equality:

$$A = A_I H = AWH \tag{65}$$

then, (65) can be expanded as follows:

$$\begin{cases} f_1 \cong AWH_{:,1} = H_{1,1}f_{j1} + H_{2,1}f_{j2} + \ldots + H_{k,1}f_{jk} \\ \quad\quad\quad\

function obtaining such that $\|E-AW+GF^T\|_F^2+Tr(\varphi^T(E-AW+GF^T))=\frac{\mu}{2}\left\|E-AW+GF^T-\frac{\varphi}{\mu}\right\|_F^2$. Then, (74) can be formulated as follows:

$$\min_W \alpha Tr\left(W^TA^TLAW\right)+\lambda Tr\left(W^TQW\right)+\frac{\mu}{2}\left\|E-AW+GF^T-\frac{\varphi}{\mu}\right\|_F^2 \tag{75}$$

where φ is Lagrange multiplier. F can be updated as follows:

$$\min_F \frac{\mu}{2}\left\|E-AW+GF^T-\frac{\varphi}{\mu}\right\|_F^2 \; s.t \;\; F^TF=I \tag{76}$$

Optimization problem (76) can be analytically solved. If we assume $Z=AW-E+\frac{\varphi}{\mu}$, $\frac{\mu}{2}\|GF^T-Z\|_F^2$, then (76) optimization problem can be reformulated as following:

$$\min_F \|GF^T-Z\|_F^2=\min_F Tr\left(\left(GF^T-Z\right)^T\left(GF^T-Z\right)\right)=$$

$$\min_F Tr\left(FG^TGF^T\right)-2Tr\left(FGZ^T\right)+Tr\left(Z^TZ\right)=\max_F Tr\left(F^TG^TZ\right)$$

$$s.t \quad F^TF=I \tag{79}$$

where $FG^TGF^T=I$ and $G^TG=I$ and $Tr(Z^TZ)$ can be ignored because it does not have F term. G^TZ can be decomposed, $SVD(G^TZ)=U\Sigma V^T$, and result is $\max_F Tr(F^TG^TZ)=\max_F Tr(F^TU\Sigma V^T)$. By introducing $K=VF^TU$, we can have $\max_F Tr(F^TG^TZ)=\max_F(K\Sigma)$. Since the K is orthogonal and no elements of K can take value more than 1, maximum value for $\max_F(K\Sigma)$ can be achieved if K be identity matrix. Consequently, $F=UKV^T=UIV^T$ is analytic solution of (79) such that U and V are left-singular vectors and right-singular vectors of matrix $G^TZ=G^T\left(AW-E+\frac{\varphi}{\mu}\right)$.

G and H can be updated as inner iterative algorithm.

$$\min_{G,H}\frac{\mu}{2}\left\|E-AW+GF^T-\frac{\varphi}{\mu}\right\|_F^2+\gamma\|G-H\|_F^2 \; s.t \; G^TG=I, H>0 \tag{80}$$

Updating G:

$$\min_{G,H}\frac{\mu}{2}\left\|E-AW+GF^T-\frac{\varphi}{\mu}\right\|_F^2+\gamma\|G-H\|_F^2 \quad s.t \;\; G^TG=I \tag{81}$$

Updating H:

$$\min_H \; \gamma\|G-H\|_F^2 \; s.t \; H>0 \tag{82}$$

E is the last parameter must be updated:

$$\min_E \|E\|_{2,1}+\frac{\mu}{2}\|E-AW+GF^T\|_F^2 \tag{83}$$

The algorithm of this feature selection method is illustrated in Algorithm 3.

Algorithm 3. Feature selection with Orthogonality constraint.

Input: Data matrix A, parameters γ, μ, λ and α
Initialize E, W, G, F, H, ϕ
repeat
  repeat
    update G, (70)
    update H, (71)
  until convergence
  update W, (66)
  update F based on analytical solution
  update E, (72)
  update Lagrange multiplier $\phi \leftarrow \phi+\mu(E-AW+GF^T)$
until convergence

After convergence, matrix W is ranked based on Euclidean norm of each row which shows the importance of each feature. In term of computation complexity, there are two loops inside each other, it means in each iteration G and H are updated which increase the time complexity. This technique has four parameters which must be tuned to obtain the best performance of algorithm.

Construction of the similarity matrix significantly affects on the performance of feature selection in graph-based feature selection techniques. Most of methods consider the fixed graph and remain it constant in the optimization iterations. The subtle point must be noted is that the constructed graph on raw data could be contaminated to noise and outlier which this leads to degradation of feature selection algorithm. To deal with this problem, an unsupervised feature selection via adaptive graph learning and constraint (EGCFS) was proposed to select most discriminative features (Zhang et al., 2022). This technique embedded graph constraint in the objective function to provide a connection among data points with closer distance, and secondly, construction of the similarity matrix embedded into optimization function to have adaptive graph by updating in each iteration. In addition, between-class scatter matrix is integrated to adaptive graph structure to improve the performance of feature selection. The objective function of this technique is constructed based on the low-dimensional manifold and Laplacian graph as follows:

$$\min_W Tr\left(W^TALA^TW\right) \;\; s.t \;\; W^TW=I \tag{84}$$

where W, A and L are orthogonal projection matrix, data and Laplacian matrix, respectively. Laplacian matrix is defined as $L=D-S$, where D is degree diagonal matrix and S is similarity symmetric matrix. Two terms are added to (84) to maximize between-class scatter matrix in the low-dimensional manifold and sparsify the solution. $S^T\mathbf{1}=\lambda\mathbf{1}$ as a constraint is applied on the similarity matrix to make connection for nearer data points.

$$\min_{W,G,S_\gamma} Tr\left(W^TALA^TW\right)+\gamma\|S\|_F^2-\lambda Tr\left(W^TAG\left(G^TG\right)^{-1}G^TA^TW\right)+\alpha\|W\|_{2,1}$$

$$s.t \;\; W^TW=I, S^T\mathbf{1}=\lambda\mathbf{1} \tag{85}$$

where $G\in\{0,1\}$ is the binary index matrix and γ, α and λ are coefficient parameters. The main difference between γ and λ is that γ is adaptively updated during the optimization process.

In another study, an adaptive graph-based unsupervised feature selection is proposed to tackle imbalanced neighbors' defect (Huang and Yang, 2022). In this technique, discriminative features are selected based on the adaptive graph and dependency score (AGDS) theories. In AGDS, K-nearest neighbors for each sample are adaptively obtained and dependency score to measure uncertainty of each feature is calculated by integrating mutual information and entropy. The objective function of AGDS, which is developed based on the adaptive graph feature selection, can be formulated as follows:

$$\min_{W,s}\sum_{ij}^{m}\left\|W^Ta_i-W^Ta_j\right\|_2^2 s_{ij}^{\;t}+\alpha\|W\|_{2,1}$$

$$+\beta\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left(H(f_j)\left\|w^j\right\|_2+H(f_i)\left\|w^j\right\|_2\right)MI(f_i;f_j)$$

$$s.t\sum_j s_{ij}=1, s_{ij}\geq 0, \left\|s^i\right\|_{2,0}=k, t>1 \tag{86}$$

where $H(f_j)$ and $MI(f_i;f_j)$ are entropy of ith feature to show the uncertainty and mutual information between i^{th} and j^{th}, respectively. Likewise, in order to prevent and mitigate imbalanced neighbor defect $\left\|s^i\right\|_{2,0}=k$ is considered as constraint to have exact $k-$ nearest neigh-

bors.

Haung et al. employed regression-based embedded feature selection followed by $l_{2,1}$ sparse regularization term and adaptive graph structure learning (Huang et al., 2021). This technique is unsupervised and considers the label as a parameter, which is learned in optimization process based on the supervised regression method, in order to obtain labels information in the unsupervised scenario. Additionally, extended total scatter matrix is considered as a constraint to reduce feature redundancy. The objective function can be expressed as follows:

$$\min_{W,F,b}\left\|A^TW+\mathbf{1}b^T-F\right\|_F^2+\lambda\|W\|_{2,1}\ \text{s.t}\ W^TR_t^{'}W=I,F^TF=I \tag{87}$$

where $R_t^{'}=R_t+\gamma D_w+\alpha ALA^T$ is extended uncorrelated constraint and R_t is total scatter matrix and D_w is a diagonal matrix to preserve the $R_t^{'}$ non-singular. F plays role of label and is learned in optimization process.

In general, feature selection techniques based on subspace learning consists of a loss function, which measures distance between original space and subspace, and regularization terms. Different loss functions and regularizations are summarized in Table 2.

Feature selection using subspace learning can take different regularization including sparse regularization and structure learning regularization. Fig. 3 shows the taxonomy of feature selection using subspace learning with different regularization functions.

Using low rank properties of data can improve the performance of feature selection. Nuclear norm is frequently used as regularization to approximate low rank property of data. In a feature selection study, Schatten p-norm ($0<p<1$) regularization as rank constraint is applied to exploit inherent rank property of data (Wang et al., 2022a).

The Schatten p-norm is defined as $\|A\|_{Sp}=(\sum_{i=1}^{\min\text{CHECKCHARACTERNOTSUPPORTED}\square\{n,m\}}\sigma_i^p)^{\frac{1}{p}}$, where σ_i is i-th singular value of matrix A. Then, the feature selection based on Schatten p-norm is expressed as follows:

$$\min_W\|A-AW\|_F^2+\lambda_1\|W\|_{2,1}+\lambda_2\|W\|_{sp}^p \tag{88}$$

where λ_1 and λ_2 are regularization coefficients. To solve (88), an auxiliary variable Q is introduced and Lagrange function with Lagrange multiplier Λ for (88) can be expressed as follows:

$$L(W,Q,\Lambda)=\|A-AW\|_F^2+\lambda_1\|W\|_{2,1}+\lambda_2\|Q\|_{sp}^p+<\Lambda,W-Q>+\lambda_3\|W-Q\|_F^2 \tag{89}$$

Equation (89) can be solved using alternating direction method of multipliers (ADMM) technique.

Matrix properties which are deeply rooted in linear algebra can be utilized for dimensionality reduction and feature selection. Obtaining bases of data which can span whole data can be directly applied for feature selection. To this end, A technique for feature selection was introduced, which relies on the construction of a matrix basis (Saberi-Movahed1 et al., 2020). In this technique distance between basis of feature matrix A and subset of A^J is minimized as follows:

$$arg\min_J distanc\left(B,A^J\right) \tag{90}$$

$$s.t\ |J|=k$$

where B is the basis matrix for A.

The Equation (90) objective function can be formulated like NMF as follows:

$$\min_{P,H}\frac{1}{2}\left\|B-A^JP\right\|_F^2 \tag{91}$$

Therefore, we can imply that $A^J\approx BH$ by taking NMF theorem.

$$\min_{P,H}\frac{1}{2}\|B-\mathrm{BH}P\|_F^2 \tag{92}$$

This optimization problem can be solved for P and H separately using alternating process. The matrix B must be constructed before iterative loop of feature selection algorithm. To this end, all the features are ranked based on information gain (IG) in descending order. The first feature of sorted matrix A is considered as first basis vector and searching for basis vectors is started from the second feature to complete basis matrix. Algorithm 4 shows the feature selection by basis matrix construction.

Algorithm 4. Algorithm of basis matrix construction

Input: $A\in R^{n\times m}$, $r=rank\ (A)$
Output: Basis matrix $B\in R^{n\times r}$
1-find IG (A)
2- A^{sorted}=sorted A based on IG in descending order
3-set $B=\{A_{:1}^{sorted}\}$, $A_{:1}^{sorted}$is first column of A^{sorted}
for i=2:m
  if $A_{:i}^{sorted}\notin span(B)$
    $B\leftarrow B\cup\{A_{:1}^{sorted}\}$
  end if
  if $rank\ (B)=r$
    break
  end if
end for

Finally, the norm of rows for matrix H after convergence are calculated and sorted in descending order, and k top features are selected. The total time complexity of this technique is $O(m+n^2k)$. It is important to note a subtle point: before applying the basis matrix construction, matrix A must be sorted. This is because while the dimension of the column space of a matrix is independent of the order, the order of the features (columns) is significant for the feature selection task. Basis matrix B would be a full rank matrix and a subspace for matrix A.

The columns of basis matrix B represent the independent columns of matrix A. This theory can be seen in another study which propose a feature selection technique based on finding independent columns of matrix A using reduced row Echelon (RRE) form (Ebrahimpour et al., 2017). RRE is an established method in linear algebra in order to obtain columns which are linearly independent. Although the column space of matrix A is independent of columns order, independent columns of matrix A can be different for different order of columns. To this end, the IG is calculated for each feature and columns of matrix A sorted based on

Table 2
Different types of regularizations and distance metrics.

Distance metric	Sparse regularization	Local learning regularization	Redundancy regularization
$\|span(A)-span(A_J)\|_F$	$\|W\|_{2,1}$	Fixed graph learning: $\sum_{i,j}(\|W^Ta_i-W^Ta_j\|_2^2S_{ij}$	Correlation among the features: $\mathbf{1}_{k\times k}^TW^TA^TAW\mathbf{1}_{k\times k}$
$\|span(A)-span(A_J)\|_{2,1}$	$\|W\|_{2,p}$, $0\le p<1$	Adaptive graph learning: $\sum_{i,j}(\|W^Tx_i-W^Tx_j\|_2^2s_{ij}+\alpha {s_{ij}}^2$	
	$\|W\|_{2,1-2}$		
$\|W\|_{2,0}$	$\sum_{i,j=1,i\neq j}^d<w^i,w^j>$		

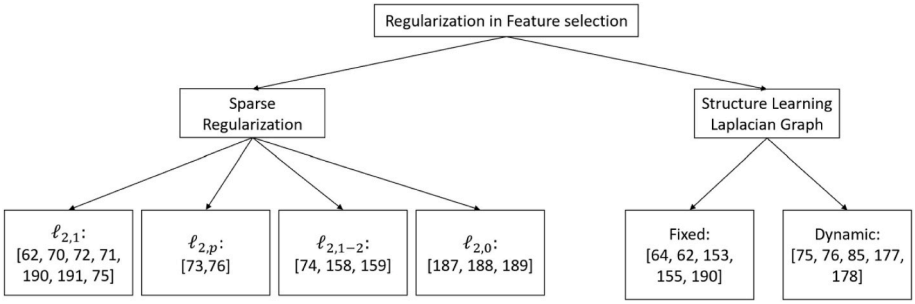


Fig. 3. The hierarchical structure of feature selection techniques based on the taxonomy of regularization functions.

IG in descending order. Selected features were independent columns of matrix A after applying RRE.

In Saberi-Movahed1 et al. (2020) and Ebrahimpour et al. (2017), the goal is to find a subspace (basis matrix) to span original data. Likewise, a feature selection technique based on the matrix factorization and singular value decomposition (SVD) is proposed (Zare et al., 2019). In this method, SVD is applied to find the orthonormal basis for matrix feature A. $SVD(A) = U\Sigma V^T$ decomposes the matrix to three matrices such that U and V are orthonormal matrices and Σ is a diagonal matrix whose elements are singular values of matrix. The columns of U and V are eigenvectors of A^TA and AA^T , respectively. The decomposition is as follows:

$$A = [u_1 \ldots u_r | u_{r+1} \ldots u_m] \begin{bmatrix} \sigma_1 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 \\ 0 & 0 & \sigma_r & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \ldots v_r | v_{r+1} \ldots v_n \end{bmatrix} \quad (93)$$

The first r ($r = rank\ (A)$) columns of U are another representation of A. $\mathscr{C}(A) = [u_1\ u_2 \ldots u_r]$ is column space of matrix A and $\mathscr{N}(A) = [u_{r+1} \ldots u_m]$ is null space of matrix A. $\mathscr{C}(A)$ is a subspace matrix which can span matrix A. Therefore, NMF feature selection formula can be rewritten as follows:

$$\min_J \left\| A - A^J \Sigma V^T \right\|_F \quad s.t \quad |J| = k \quad (94)$$

Based on NMF feature selection formulation, $H = \Sigma V^T$, $A^J = AW \approx U$ and $V^T \approx (A^J\Sigma)^{\dagger} A$ can be considered, where $(A^J\Sigma)^{\dagger}$ is the Moore-Penrose pseudoinverse of $A^J\Sigma$. The information gain (IG) of matrix A is calculated and added to cost function to maximize the relevancy. Therefore, $\|Wg\|_F$ is IG function which must be optimized in terms of W, where W is indicator matrix and g is a vector whose elements are IG of each feature. Finally, the objective function for supervised matrix factorization SVD-based technique is defined as follows:

$$\min_W \frac{1}{2}\|A - AWH\|_F^2 + \frac{\beta}{4}\|W^TW - I_k\|_F^2 - \frac{1}{2}\|Wg\|_F^2 \quad s.t \quad W \geq 0 \quad (95)$$

Solving this optimization problem is tough task, $W \geq 0$ is converted to $Tr(\gamma W)$ and added to objective function using Lagrange technique. Where γ is the Lagrange multiplier. W is placed to iterative loop like NMF feature selection to find top k features. In another study, QR

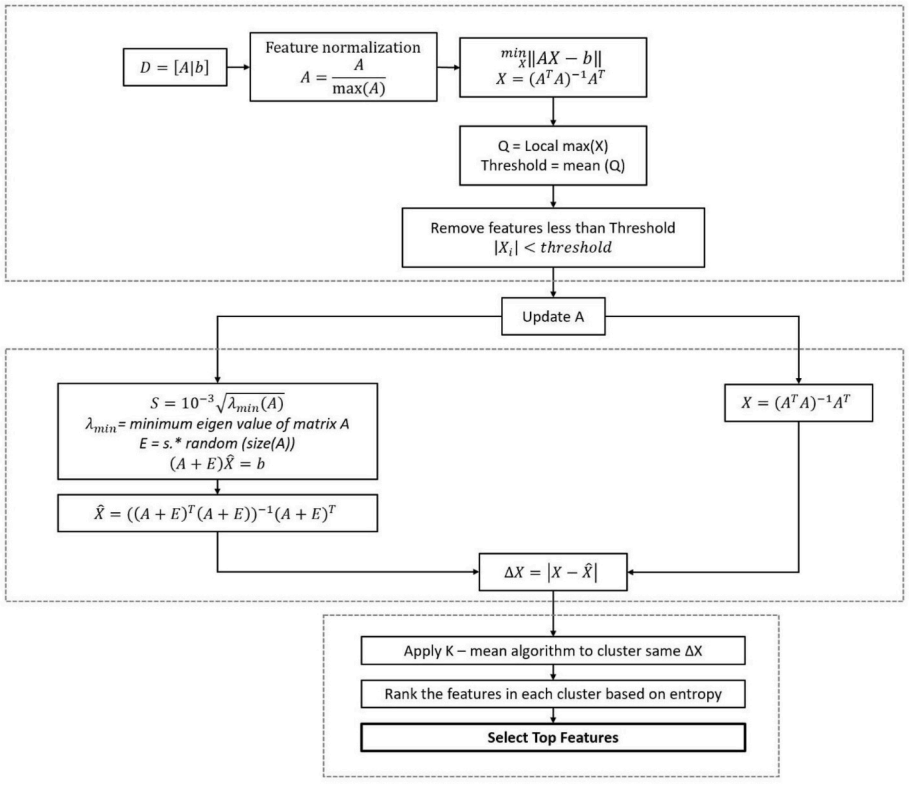


Fig. 4. Schematic of feature selection using perturbation theory. This figure illustrates three steps of feature selection using perturbation theory including filtering out features under threshold, comparing with perturbed data and clustering by k-mean feature selection.

decomposition and NMF are combined for feature selection (Jahani Samareh et al., 2023). QR method factorizes a matrix to an orthogonal matrix and upper triangle matrix. In this feature selection technique, $span(A)$ is approximated by orthogonal matrix of QR and $\|Q - AWH\|_F^2$ is considered as loss function where Q is orthogonal matrix resulted by QR decomposition. Correlation of samples and features are considered as regularization to minimize redundancy among the features and among the samples.

In the context of matrix properties for feature selection, perturbation theory is applied to find top features with supervised approach (Afshar and Usefi, 2020). In this technique, the least square problem is solved for matrix A and perturbed matrix $A + E$, where E is a random matrix with $\|E\|_2 = \sigma_{min}(A)$. The solution of least square problem of matrix A and perturbed matrix $A + E$ are subtracted to calculate $\Delta x = \|Ax - b\| - \|(A + E)x^{'} - b\|$. $\Delta x = x - x^{'}$ is calculated for each feature and features with similar Δx are grouped by clustering technique such as K-mean. Features in each cluster are ranked based on entropy and first rank of each cluster is considered as top feature. Fig. 4 illustrates the flow of feature selection based on perturbation theory. This is possible to have some big clusters contain independent features. To tackle this challenge, the angle between features and target are calculated to avoid multi-collinearity (Anaraki and Usefi, 2019). The theory of feature selection using perturbation theory can be seen in the context of null-space analysis, condition number of matrix and matrix sensitivity. Saber-i-Movahed1 et al. (2020), Ebrahimpour et al. (2017), Zare et al. (2019), Afshar and Usefi (2020) and Anaraki and Usefi (2019) utilized the matrix properties to obtain the column space of matrix A which has independent columns. The selected features are determined based on the indices of these independent columns. Null space analysis can be useful to interpret the features and obtain most discriminative features. Likewise, Usefi et al. (Afshar and Usefi, 2021) utilized null space to extract singular vectors of high-dimensional data to cluster same features and remove the redundant features.

Utilizing set theory is another approach for feature selection. Mokhtia et al. combined hesitant fuzzy set (HFS) theory and dual manifold in order to obtain the most important features using regression model (Mokhtia et al., 2021). In this technique, regression cost function with different regularization functions including; Ridge, Lasso and Elastic-Net is utilized such that feature manifold regularization term and inter-class samples manifold regularization term are added to preserve the geometry information in the data space and feature space. In this technique, hesitant fuzzy correlation matrix (HFCM) is calculated for features and inter-class samples, and Laplacian matrix of the feature graph and Laplacian matrix of the inter-class samples graph are computed as well.

These steps must be performed for HFCM construction; 1) fuzzy C-mean is applied to cluster samples, 2) the obtained cluster memberships are projected to the associated features to acquire HFSs over the features. This procedure can be repeated for samples to calculate HFSs over the samples. 3) finally, the HFCM is constructed by calculating HFC among HFSs. Γ^F and Γ^S are HFCM among features and samples, respectively. Diagonal matrices $D_{ii}^F = \sum_{j=1}^{n} \Gamma_{ij}^F$ and $(D_k^S)_{ii} = \sum_{j=1}^{p} (\Gamma_k^S)_{ij}$ are defined to calculate Laplacian matrix such that n, k and p are the number of features, the class of samples and the number of samples belonging to the kth class, respectively. Consequently, $L^F = D^F - \Gamma^F$ and $L_k^S = D_k^S - \Gamma_k^S$ as feature Laplacian matrix and sample Laplacian matrix are obtained, respectively. Therefore, the cost function for feature selection using dual-manifold regularization and HFC with different regression regularizations can be formulated as follows:

$$F(W) = \frac{1}{2}\|Y - AW\|_2^2 + R(W) + \frac{\beta}{2}\left(W^T L^F W\right) + \sum_{k=1}^{C} \frac{\upsilon_k}{2}\left(W^T A_k^T L_k^S A_k W\right) \quad (96)$$

where β and υ_k are regularization hyper parameters, C is the number of classes and Y is label vector. $R(W)$ is regularization term to keep balance between error of fitting and complexity of model. Different $R(W)$ are considered for this technique including Ridge regression $R(W) = \frac{\alpha}{2}\|W\|_2^2$, Lasso regression $R(W) = \frac{\alpha}{2}\|W\|_1$ and Elastic-Net regression $R(w) = \lambda\left(\frac{\alpha}{2}\|W\|_2^2 + \frac{1-\alpha}{2}\|W\|_1\right)$. λ and α are hyper parameters to control error of fitting and sparsity, respectively.

Although Equation (96) with Ridge regression can be solved by taking derivative of $F(W)$ with respect to W, Equation (96) with Lasso and Elastic-Net should be solved as non-smooth problem since L1 norm is not Lipschitz Continuous. Details and mathematics of HFCM and HFS can be found in Rodríguez et al. (2014) and Chen et al. (2013). Several studies utilized the rough-set and Fuzzy-set for feature selection (Mokhtia et al., 2020; Eftekhari et al., 2022; Mafarja et al., 2019; Wang et al., 2019; Sun et al., 2020). In Sun et al. (2020), fuzzy neighborhood multi-granulation rough sets (FNMRS) is utilized to find the most informative features. In this technique, FNMRS is applied to construct uncertainty measures using fuzzy neighborhood rough sets concept.

Various distance metrics can be utilized to extract a subspace that captures the maximum information from the original space. Euclidean and Hausdorff distances, principal angles, Grassmann and Chordal distances, Symmetric and Directional distances, and Variance-Covariance Subspace Distance (VCSD) are metrics which can be utilized for subspace learning (Karami et al., 2023). Karami et al. (2023) proposed an unsupervised feature selection based on Variance-Covariance. This technique, which is called Variance-Covariance Subspace Distance Feature Selection (VCSDFS), capitalizes on the information correlation among data features, enabling the identification of feature subsets that possess the minimum norm property in their corresponding Variance-Covariance matrix (VCSD $= min \left\|AA^T - A_J A_J{}^T\right\|_F^2 = \min_W \left\|AA^T - AWW^T A^T\right\|_F^2$, where A and W are data matrix and projection matrix).

Table 3 illustrates the categories of feature selection using matrix factorization. Techniques are categorized based on loss function, sparsity regularization, structure learning regularization, minimum redundancy regularization, low-rank property consideration and label information.

2.1. Computational complexity

Computation complexity of algorithm must be computed for each algorithm. For instance, Algorithm 1 (Wang et al., 2015a) with input matrix data $\in R^{n \times m}$, where n is the number of samples and m is the number of features, has $O(k^2m + nm^2 + km^2)$ time complexity for each iteration. Where k is the selected features. In Algorithm 2 (Saber-i-Movahed et al., 2022a), time complexity for updating W and H in each iteration are $O(2nm^2 + 2km^2 + 4k^2m)$ and $O(4k^2d)$, respectively. Since k is fixed parameter and $m \gg n$, we have $O(m^2)$. Table 4 shows the time complexity of feature selection techniques.

3. Sparse representation learning for feature selection: compressed sensing and dictionary learning

Compressed sensing (CS) is a technique that can be applied to reconstruct a signal with sampling rate smaller than Nyquist sampling rate (Eldar and Kutyniok, 2012). Recently, this theory has been leveraged for feature selection. Conrad et al. (2017) proposed a new feature selection based on CS for feature selection and Xu et al. (2020) developed this technique to increase robustness against noise.

Based on CS theory, signal must be sufficiently sparse, which is considered as prior information, for recovering signal accurately. Since signal in time domain is not often sufficiently sparse, the sparse representation of signal can be achieved by Fourier transform, discrete cosine transform and wavelet transform. The measurement step is applied after getting sparse representation such that a measurement matrix is applied on sparse signal to reduce the dimension (Luo et al., 2022). Signal

Table 3
The category of matrix factorization-based feature selection based on different approaches.

Methods	Loss function	Sparse Regularization (Global learning)	Geometrical information preservation (Local learning)	Minimum Redundancy	Low-rank property	Dual space Consideration	Label Information
Wang et al. (2015a)	$\|.\|_F$	–	–	–	–	–	Unsupervised
Wang et al. (2015b)	$\|.\|_F$	–	–	Correlation	–	–	Unsupervised
Wang et al. (2020)	$\|.\|_F$	–	Fixed graph	–		–	Unsupervised
Wang and Wang (2017)	$\|.\|_F$	$\|.\|_{2,1}$	Fixed graph	–	–	–	Unsupervised
Nie et al. (2010)	$\|.\|_{2,1}$	$\|.\|_{2,1}$	–	–	–	–	Supervised
Nie et al. (2016)	–	$\|.\|_{2,1}$	Adaptive graph	–	–	–	Unsupervised
Shang et al. (2020a)	$\|.\|_F$	$\|.\|_{2,1/2}$	Adaptive graph	–	–	–	Unsupervised
Shi et al. (2018)	$\|.\|_F$	$\|.\|_{2,1-2}$	–	–	–	–	Unsupervised and Supervised
Meng et al. (2018)	$\|.\|_F$	$\|.\|_{2,1}$	Fixed graph	–	–	Local learning	Unsupervised
Saberi-Movahed et al. (2022a)	$\|.\|_F$	Inner-product	–	Correlation	–	Global learning	Unsupervised
Qi et al. (2018)	$\|.\|_F$	Inner-product	–	–	–	–	Unsupervised
Shang et al. (2020b)	$\|.\|_{2,1}$	$\|.\|_{2,1}$	–	–	Nuclear norm	–	Unsupervised
Hu et al. (2022)	$\|.\|_F$	$\|.\|_{2,1}$	Fixed graph	–	–	Local learning	Unsupervised
Luo et al. (2022)	$\|.\|_{2,1}$	$\|.\|_{2,1}$	Fixed graph	–	–		Unsupervised
Huang et al. (2021)	$\|.\|_F$	$\|.\|_{2,1}$	–	uncorrelated constraint	–	–	Unsupervised
Wang et al. (2022a)	$\|.\|_F$	$\|.\|_{2,1}$	–	–	Schatten p-norm	–	Unsupervised
Mokhtia et al. (2021)	$\|.\|_2$	$\|.\|_1$, $\|.\|_2$ and $\|.\|_2^2 + \alpha\|.\|_1$	Fixed graph	Hesitant fuzzy correlation matrix	–	–	Supervised
Karami et al. (2023)	$\|.\|_F$	Inner-product	–	–	–	–	Unsupervised

Table 4
Time complexity of each feature selection technique ($n \gg k, m \gg n$).

Technique	Complexity
Wang et al. (2015a)	$O(m^2)$
Wang et al. (2015b)	$O(m^2)$
Wang et al. (2020)	$O(m^2)$
Wang and Wang (2017)	$O(m^2)$
Tang et al. (2019)	$O(m^3)$
Shang et al. (2021)	$O(m^2)$
Zhang et al. (2019b)	$O(m^3)$
Miao et al. (2021)	$O(m^3)$
Nie et al. (2020)	$O(m^2)$
Meng et al. (2018)	$O(m^2)$
Saberi-Movahed et al. (2022a)	$O(m^2)$
Shang et al. (2020b)	$O(m^3)$
Hu et al. (2022)	$O(m^2)$
Zhang et al. (2022)	$O(m^3)$
Huang and Yang (2022)	$O(m^3)$
Huang et al. (2021)	$O(m^3)$
Saberi-Movahed1 et al. (2020)	$O(m^2 + n)$

recovery based on CS can be expressed as follow:

$$\min \|S\|_0 \quad s.t \ \ \Phi \times S = y \tag{97}$$

Where S, Φ and y are sparse recovered signal, measurement matrix and compressed measurement, respectively. $\|.\|_0$ is pseudo norm that is counting the number of nonzero elements of vector. One-bit CS is leveraged to solve feature selection problem (Eldar and Kutyniok, 2012). One-bit CS is one of the quantized CS techniques which output is the sign of compressed samples as shown in (98).

$$y = \text{sign}(\Phi \times S) \tag{98}$$

Feature selection can be modeled as one-bit CS such that measurement matrix, label vector and recovered signal are replaced by feature matrix, sign information weighting coefficient vector, respectively. Weighting coefficient vector indicates the importance of each feature. The theoretical background of feature selection using CS can be found in Genzel and Kutyniok (2016). Consequently, CS feature selection as supervised feature selection technique can be expressed as follows:

$$\underset{w}{\text{argmin}} \|[y \odot AW]_-\|_1 \quad s.t \ \ \|W\|_0 = k, \|W\|_2 = 1 \tag{99}$$

Where y, A and W are label, data matrix and weight (importance of each feature) vector, respectively. $\odot$ indicates Hadamard product (element-wise multiplication), k is the number of selected features and $[.]_-$ indicates the negative function and can be implied as follows:

$$[w]_- = \begin{cases} w & if \ w < 0 \\ 0 & owtherwise \end{cases} \tag{100}$$

Conrad et al. (2017) highly suggested to perform noise reduction before using CS feature selection. To this end, it applied gaussian filter to

smooth the data. Finally, hard threshold is applied to output of (99) in order to eliminate all the element of w that are close to zero. Xu et al. (2020) developed CS feature selection technique for noisy data. In this method noise was added to feature matrix ($D = A + E$). Where E is added noise to original data matrix and $\underset{w}{\operatorname{argmin}}\|[y \odot DW]_-\|_1$ is solved for feature selection using a sub-gradient technique. In the last step, non-zero element in vector w shows the selected feature.

Dictionary learning is another approach of sparse representation learning which can be used in feature selection. The dictionary learning problem can be expressed as follows:

$$\min_{U,H}\|A - UH\|_F^2 + \lambda R(H) \tag{101}$$

Where U, H and λ are the dictionary, representation coefficients matrix and regularization coefficient, respectively. Zhu et al. (2016) proposed a feature selection technique based on dictionary learning by converting (101) into feature selection problem using $H = W^T A$ consideration, where W can be considered as predefined analysis dictionary. Therefore, feature selection problem using dictionary learning can be expressed as follows:

$$\min_{U,H,W}\|A - UH\|_F^2 + \beta\left\|H - W^T A\right\|_F^2 + \lambda\|V\|_{2,p}^p \;\; s.t \;\; \|u_i\|_2^2 \le 1, i = 1, 2, \ldots, k \tag{102}$$

Where u_i is the i-th atom of the synthesis dictionary U and β is a coefficient.

In the context of dictionary learning, Parsa et al. (2022) proposed feature selection learning using low rank dictionary learning. $\|.\|_{2,1}^2$ and Laplacian graph are applied to extract low rank property and manifold geometrical information preservation, respectively. More dictionary learning-based feature selection can be found in Miao et al. (2022), Foroughi (2015) and Li et al. (2023).

4. Information theory-based feature selection methods

Relief algorithm is the one of the classical techniques which is extensively utilized in feature selection task. The initial Relief algorithm was proposed for only binary problem (Kira and Rendell, 1992). In this technique, m samples of data are randomly selected, and for each sample the nearest neighbors (NN) of same class and opposite class are computed. As a result, the importance of features is evaluated by measuring the quality of each feature in terms of differentiation between same class and opposite class. Therefore, a feature can be considered as a good discriminator if it can discriminate two near samples that belong to different labels (Robnik-Sikonja and Kononenko, 2003). Relief technique is improved for incomplete noisy and multi-classes data by ReliefF technique (Kononenko, 1994). In this technique, k-NN of each sample for same class and other classes are assigned by ReliefF, and the vector of importance is constructed based on the interaction of features in the range of [−1,1] such that 1 shows the most important feature. ReliefF multi-labels (ReliefF -ML) is proposed for multi-classes feature selection task (Reyes et al., 2015). In this feature selection technique, prior probability of class l is calculated as follows:

$$P_l = \frac{C_l + b}{n + 2b} \tag{103}$$

where C_l, b and n are the number of samples in the data which belong to class l, smoothing parameter to control the uniformity of prior probability and the sample size, respectively.

Prior probability for same-labels (Hits) and opposite-labels (Misses) are defined as follows:

$$P_{H_i^l} = \frac{\sum_{j \in H_i^l} d_L(i,j)}{k} \tag{104}$$

$$P_{M_i^l} = \frac{\sum_{j \in M_i^l} d_L(i,j)}{k} \tag{105}$$

where $P_{H_i^l}$ is k-nearest neighbors which have the relevant label l of sample i as relevant label, and $P_{M_i^l}$ is k-nearest neighbors which have the irrelevant label l of sample i as irrelevant label. Dependency of classes is measured by $P_{H_i^l}$ and $P_{M_i^l}$, and $d_L(i,j)$ is hamming distance to measure the distance among the sets of i and j , and it is defined as follows:

$$d_L(i,j) = \frac{1}{n}\sum_{i=1}^{n}\frac{|y_i \Delta y_j|}{Q} \tag{106}$$

where y and Q are the class of sample and the number of classes. Δ indicates the symmetric difference between two sets and Hamming distance (loss) assesses the number of misclassifications for an instance (Zhang and Zhou, 2007).

Consequently, the features can be sorted (ranked) using following iterative formula.

$$W^{t+1} = W - \sum_{l \in y_i}\left(\frac{P_l}{\sum_{q \in y_i} P_q}\frac{1 - P_{H_i^l}}{1 + P_{H_i^l}}\sum_{j \in H_i^l}\frac{\delta(x_{if}, x_{jf})}{mk}\right) + \sum_{l \in \overline{y_i}}\left(\frac{P_l}{\sum_{q \in \overline{y_i}} P_q} P_{M_i^l}\sum_{j \in M_i^l}\frac{\delta(x_{if}, x_{jf})}{mk}\right) \tag{107}$$

where δ calculates the difference between sample i -feature f and sample j -feature f, and parameter m is the number of selected samples for training data. $\frac{P_l}{\sum_{q \in y_i} P_q}\frac{1 - P_{H_i^l}}{1 + P_{H_i^l}}$ and $\frac{P_l}{\sum_{q \in \overline{y_i}} P_q} P_{M_i^l}$ measure the relevancy and irrelevancy of labels in each iteration, respectively. The number of samples, as training set, and the number of nearest neighbors are input of this algorithm and W is output of this algorithm which shows the importance of each feature.

New ReliefF technique for multi-class feature selection, which is called PPT-ReliefF, is proposed to enhance the ability of traditional Relief. The Pruned Problem Transformation method (PPT) converts the original labels into new version of labels and a threshold is defined in PPT to select labels to generate new labels. Labels that occur more than threshold are considered as selected and those less than threshold are ignored. Generally, PPT technique is extended version of ReliefF which has two actions; 1) pruning and 2) subsampling. Pruning means the removing of labels less than threshold and subsampling is applied to minimize the information loss (Read, 2008).

ReliefF-ML and PPT-ReliefF are categorized as problem transformation method. These types of techniques map the labels to one or many subsets and rank the features in each subset based on the predefined score. Finally, the results of all subsets are collected and overall ranking of features can be applied to select top features.

Label correlation analysis plays an important role for multi-label feature selection. Since the problem transformation methods, such as ReliefF, have extra computation to modify data distribution, a mutual information (MI)-based technique for multi-label feature selection is proposed (Gonzalez-Lopez et al., 2019). MI for a feature f_i and two classes $Y = \{y_1, y_2\}$ can be defined as follows:

$$MI(f_i, Y) = MI(f_i, y_1|y_2) + MI(f_i, y_2|y_1) + 2 \times MI(f_i, y_1, y_2) \tag{108}$$

In this technique, vectorized form of MI is applied to compensate of MI (108) weakness, since it cannot discriminate between features with small or huge variation of information for labels. Vectorized MI can be formulated as follows:

$$MI_{Vec}(f_i, Y) = \{MI(f_i, Y)|y \in Y\} \tag{109}$$

where MI_{Vec} is a vector, with length equal to the number of labels, whose elements indicate the MI value between the feature and specific label. Consequently, two metrics based on Euclidian norm maximization

(ENM) and geometric mean maximization (GMM) were leveraged for feature selection.

$$ENM(MI_{Vec}(f_i,Y)) = \left(\sum_{j=1}^{q} MI_{Vec}(f_i,y_j)^2\right)^{\frac{1}{2}} \tag{110}$$

$$NMM(MI_{Vec}(f_i,Y)) = \left(\prod_{j=1}^{q} MI_{Vec}(f_i,y_j)\right)^{\frac{1}{q}} \tag{111}$$

In the light of MI application for feature selection to have maximum correlation and minimum redundancy, granulation multi-labels feature selection method is proposed (Li et al., 2017). In this technique, the labels are categorized into several information granules based on their correlation. As a result, the features can be selected that have maximum correlation with each label information granules and minimum correlation with other selected features. This study proved that maximum correlation between the feature and the labels information granules can be estimated by solving a maximization problem. In this technique, therefore, features are ranked and the optimum number of features are selected. Maximization problem can be written as follows:

$$max\ \frac{1}{|G|}\sum_{l_j\in G} MI(x_i;l_j) - \frac{1}{|S|}\sum_{x_j\in S} MI(x_i;x_j) \tag{112}$$

where G, S and x are information granules, feature set and specific feature, respectively. The maximum relevancy and minimum redundancy are guaranteed by first term and second term of (112), respectively.

In the context of multi-label feature selection, a method based on the pareto dominance concept is proposed (Kashef and Nezamabadi-pour, 2019). This method considered multi-label feature selection as multi-objective optimization problem (MOP) and found the best features based on filter strategy. In this technique, features are mapped to subspace with dimension equal to number of labels and the most informative features are selected based on Pareto. This technique is categorized as label-specific feature selection method since the correlation between features and each label is individually explored. Although the number of top features or selected features must be determined by user as input of algorithm, in this technique the optimum number of selected features is obtained by algorithm.

This technique can be directly applied on data and data transformation is not required. Additionally, it can be applied for online feature selection. Although the MOP has a set of optimal solutions such that there are not superior to one another, Pareto can be applied to find dominant solution. Pareto optimal is considered for a feature if it is not dominated by other features. Therefore, a set of non-dominated features make Pareto optimal. Therefore, the algorithm of this technique can be summarized in 4 steps: A) calculate symmetrical uncertainty Feature label (SUFL) matrix, B) map the data to subspace with dimension equal to the number of labels based on correlation between feature and labels using SUFL, C) apply Pareto optimal to find non-dominated features and D) report non-dominated features as optimum selected features.

Symmetrical uncertainty (SU) is defined as follows:

$$SU(label,feature) = 2\times\frac{IG(label|feature)}{Entropy(label)+Entropy(feature)} \tag{113}$$

SUFL can be constructed as follows:

$$SUFL(label,feature) = \begin{bmatrix} SU(label1,feature1) & \cdots & SU(label1,feature\ \boldsymbol{m}) \\ \vdots & \ddots & \vdots \\ SU(label\ \boldsymbol{q},feature1) & \cdots & SU(label\ \boldsymbol{q},feature\ \boldsymbol{m}) \end{bmatrix} \tag{114}$$

where q and m are the number of labels and number of features, respectively.

In the context of multi-labels feature selection and MI, a technique based on combination of mutual information and convex optimization is proposed (Sun et al., 2019a,b). In this technique, correlation of labels is used to formulate the feature selection problem as convex optimization problem. Minimum redundancy and maximum relevancy are considered to find the best features. Therefore, the objective function can be expressed as follows:

$$\max_S \sum_{f_i\in S}\sum_{l_j,l_k\in L, j\neq k} MI(f_i;l_j)\times Cor_J(l_j;l_k) + \sum_{f_i\in S}\sum_{l_j\in L} MI(f_i;l_j) - \sum_{f_i,f_j\in S} MI(f_i;f_j) \tag{115}$$

where Cor_J is Jaccard correlation which measures the label correlation. S and L are selected features subset and labels, respectively. The first term of (115) considers label correlation information and maximum relevancy-minimum redundancy together. $Cor_J(l_j;l_k)$ plays role as weighting matrix for feature-label pairs. Second term and third term of equation represent relevancy and redundancy, respectively. Matrix $D_{i,j} = MI(f_i;f_j)$, vector $c_i = MI(f_i;l_j)$ and vector $e_i = \sum_{l_j\in L} MI(f_i;l_j)$ $\sum_{l_k\in L} Cor_J(l_j;l_k)$ are defined to reduce computation time. $D_{i,j}$ is a square matrix with size equal to number of features. Then, equation (115) can be reformulated as follows:

$$\max_S \sum_{f_i\in S} e_i + \sum_{f_i\in S} c_i - \sum_{f_i\in S}\sum_{f_j\in S} D_{i,j} \tag{116}$$

This formula can be expressed as a convex objective function by inspiring sparse learning based methods. Therefor, a weight vector w with length equal to the number of features is defined such that w_i indicates the weight of ith feature. Consequently, mutual information-convex objective function for multi-label feature selection can be expressed as follows:

$$\max_w c^T w + \alpha c^T w - \beta w^T D w \quad s.t\ w_i \geq 0, \sum_{i=1}^{n} w_i = 1 \tag{117}$$

where α and β are two positive parameters to control objective function terms.

It can be seen that the application of information theory in feature selection has been deeply rooted. In this vein, an information theory-based feature selection method that leverages neighborhood information theory, which is combination of shanon's information theory and neighborhood concept, is proposed (Lin et al., 2015). In this technique, all samples with different labels are granulated by defining a margin of instance. Then, three cognitive viewpoints (CV) are applied to select top features. Since the probability density of features is unknown, neighborhood information entropy and neighborhood mutual information are merged to generate new criteria. This technique works based on the maximization of following objective function:

$$J(f^+,S;L) = NMI^{\delta^{CV}}(f^+;L) - NMI^{\delta^{CV}}(f^+;S) + NH^{\delta^{CV}}(S) - NH^{\delta^{CV}}(L) \tag{118}$$

where L, S and f^+ are label set, selected feature and a new (candidate) feature, respectively. $NMI^{\delta^{CV}}(f^+;L)$ is multi-label neighborhood mutual information (NMI) of f^+ and L , and NMI can be computed as follows:

$$NMI^{\delta^{CV}}(f^+;L) = -\frac{1}{n}\sum_{i=1}^{n} log\frac{\left\|\delta^{CV}_{f^+}(x_i)\right\|\cdot\left\|\delta^{CV}_{L}(x_i)\right\|}{n\left\|\delta^{CV}_{f^+\bigcup L}(x_i)\right\|} \tag{119}$$

where x_i is a sample and $\delta_{f^+\bigcup L} = \delta_{f^+}(x_i)\cap\delta_L(x_i)$ $(\delta^{CV}(x) = \{x_i|distance(x,x_i)\leq m^{CV}\})$.

$m^{CV} = distance(x,NS(x)) - distance(x,NT(x))$ is the margin or radius to find neighborhood of granule, and also NS and NT are different label neighbors and same labels neighbors, respectively. Three different CV are surveyed in this method:

1) Optimistic viewpoint ($m^{opt}(x) = m_l(x)$ where $l = arg\max_{l_i} m_{l_i}(x)$),

2) Neutral viewpoint ($m^{nue}(x) = \frac{1}{L}\sum_{i=1}^{L} m_{l_i}(x)$)

and

3) Pessimistic viewpoint ($m^{Pes}(x) = m_l(x)$ where $l = arg\min_{l_i} m_{l_i}(x)$).

$NH^{\delta^{CV}}$ is multi-label average uncertainty (multi-label neighborhood entropy) which can be computed as follows:

$$NH^{\delta^{CV}} = -\frac{1}{n}\sum_{i=1}^{n} log\frac{\left\|\delta_{f^+}^{CV}(x_i)\right\|}{n} \tag{120}$$

In a nutshell, this transform-based technique can be summarized into three steps: A) granulate the samples by finding margin based on one the CV methods, B) exploration to find nearest different/same label neighbors for each sample and C) rank the features and select the top features.

Although achieving to the minimum number of features is the main aim of feature selection, redundancy recognition straightly is not easy. To tackle this challenge, a feature selection method based on Markov blanket (FSMB) theory is proposed (Peng et al., 2005b). Given label l_i, feature f_i and features subset $M \subset F(f_i \notin M)$. M can be considered as Markov blanket for f_i if and only if there exist $P(F - M - \{f_i\}, l_i | f_i, M) = P(F - M - \{f_i\}, l_i | M)$. The Markov blanket says that the subset M not only contains the information of feature f_i but also all the features in subset. Consequently. The best subset of features can be achieved by backward elimination. f_i is redundant feature and must be removed from current subset G if it has Markov blanket M in G and is weakly relevant (Lei and Liu, 2004). In this method, the feature space is divided to four parts: irrelevant features, weakly relevant and redundant features, weakly relevant but non−redundant features and strongly relevant features. The goal of FSMB is to find weakly relevant but non−−redundant and strongly relevant features as optimum subset. This method has two steps: A) relevancy analysis to find relevant subset by irrelevant feature elimination, and B) redundancy analysis to remove redundant feature from relevant set. In this technique, approximate Markov blanket and predominant feature as two new definitions are introduced. For two relevant features f_i and f_j, approximate Markov blanket can be formed for f_i from f_j if and only if $SU_{j,l} \geq SU_{i,l}$ and $SU_{j,i} \geq SU_{i,l}$. Where SU is symmetrical uncertainty.

A predominant feature is defined for a relevant feature if and only if it does not form the approximate Markov blanket in set. In a nutshell, this technique has an iterative algorithm such that a predominant feature is selected and all the features that form approximate Markov blanket with this are removed. Iteration will be terminated until there is no more predominant feature. Although Markov blanket selects top features using information between feature and labels, and correlation between two features, misjudged as redundant can be happened. To circumvent this problem, strong approximate Markov blanket (SAMB) method is proposed (Hua et al., 2020). This technique introduces a new condition to consider high correlation between features with aim of redundant misjudging neutralization. This technique follows approximated Markov blanket that SU is computed as follows:

$$SU(f_i, f_j) = 2\frac{H(f_i) - H(f_i|f_j)}{H(f_i) + H(f_i)} \tag{121}$$

where H represents entropy. Therefore a new redundancy definition based on strong approximated Markov blanket to avoid misjudged is introduced as follows:

$$SU(z_i, z_j) > \delta,$$

$$\delta = \max_{f_k \in S_r, f_l \in F, k \neq l} SU(z_k, z_l) \tag{122}$$

$$SU(z_i, y) \geq SU(z_j, y) \tag{123}$$

$$SU(z_i, z_j) > SU(z_j, y) \tag{124}$$

where z and y are observation vectors of feature and label, respectively. S_r is a set of features with strong relevancy. Based on this new definition of redundancy, f_i is a strong approximated Markov blanket for f_j and f_j is redundant with respect to f_i. Computation of δ for different S_r is time costly. To this end, SAMB computes $= \sum_{k=1}^{m}\sum_{l\neq k}^{m}\frac{SU(z_l, z_k)}{m(m-1)}$. The feature selection using SAMB can be summarized: A) calculate SU for each feature, B) sort the features-based value of part A, c) the first rank feature is selected, and rest of the features are sequentially checked, and D) redundant features are obtained by strong approximated Markov blanket and all redundant features are grouped to remove.

In the light of feature selection using the concept of Markov blanket, a new technique based on combination of MI and Monte Carlo (MIMC) is proposed (Manikandan and Abirami, 2021). In this technique, approximate Markov blanket and concept of MI are applied to find irrelevant and redundant features. Additionally, Monte Carlo tree search is leveraged to select most informative features. In this method, SU is substituted by MI, since SU only considers linear dependency between features, whereas MI considers both linear and non-linear. Finally, approximate blanket for redundancy recognition is defined by replacing MI instead of SU in (122), (123) and (124). This technique can be summarized in four steps, 1) all the features are sorted based MI between each feature and labels, 2) MI of each feature-label and MI of feature-feature are calculated, 3) features with MI less than predefined threshold are removed to identify the most prominent features and prominent features groups, and 4) Markov blanket is checked for all features such that a feature is considered as redundant, which is added to prominent feature group, if it forms Markov blanket. Finally, this process is repeated until no redundant feature is detected. The predominant features groups are considered as input for neighborhood search algorithm to find irrelevant features using correlation measurement. Monte Carlo Tree search is leveraged to find optimum subset of features in order to avoid misjudging as redundant feature (Michał and Koronacki, 2018).

This point should be noted that Monte Carlo tree search (MCTS) can be directly utilized for selection task (Chaudhry and Lee, 2018). MCTS takes four actions to do feature selection task. These actions are selection, expansion, simulation, and backpropagation. There are tree policy and default policy to find the best subset of features. Selection and expansion apply tree policy and simulation use default policy. A binary tree is constructed in this technique such that each node indicates each feature and has two options select or not. This tree starts from node to leaf, and the goal is to find best path. This technique is a wrapper-based feature selection, since the accuracy of classification is considered as reward to find best path of this tree. The importance of each node is calculated by upper confidence bounds for trees (UCT). UCT is computed as follows:

$$UCT_{v_j} = \max\left(Q_{v_j}\right) + C \times \sqrt{\frac{2 \times \ln\text{CHECKCHARACTERNOTSUPPORTED}\square(N_{v_i})}{N_{v_j}}} \tag{125}$$

where $\max(Q_{v_j})$ indicates maximum reward at node v_j and C is constant parameter to control balance between exploration and exploitation. N_{v_j} and N_{v_i} are the number of selected nodes (feature) j and its parent (feature) I, respectively.

Likewise, features are initialized and new features are added to create a subset of feature at each iteration by following tree policy and default policy. Each node is evaluated by (125) at each level of tree to

determine whether this feature must be selected or not. Tree is expanded by adding new feature and decision is made for this feature by (125). If the new feature is selected as important feature, which is called child, it will be added to the best path of tree (selected subset feature). Based on default policy, the path of the tree that include selected features is full randomly selected. The selected path contains some features, and these features are ranked by ReliefF and top k features are selected as candidate features for classification. The accuracy of classifier is used as reward to backpropagate on the path of tree.

5. Evolutionary computation

As aforementioned, features selection is categorized as combinatorial NP-hard problem and searching for the best features is difficult and challenging task. Random search can be considered to remedy this challenge, since examine all feasible solutions is impossible. In random search, a solution is randomly generated and it is improved by iterative approach based on assigned metric. Heuristic searches as another type of search method can be leveraged for feature selection. The metaheuristic algorithms have drawn increased attention during last decade due to the ability of solving feature selection problem in affordable time (Faris et al., 2019; Rashedi and Nezamabadi-pour, 2014). Metaheuristic algorithms can be categorized into two main groups: single-solution-based (S-Metaheuristics) and population-based (P-metaheuristics) (Talbi, 2009). However, there are other classification types for Metaheuristic algorithms based on other properties of algorithms such as swarm intelligence-based algorithms, evolutionary based algorithms, and physics-based algorithms.

Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO) and Ant Colony Optimization (ACO) are studied in this survey.

5.1. Genetic algorithm

GA is categorized as evolutionary and P-metaheuristics optimization algorithm which is inspired from Darwinian biological system principle. Exploration and exploitation are considered as main principles by GA to have local search and global search for obtaining global optimal solution. Crossover and mutation are two main actions of GA which randomly applied on chromosomes. Crossover is a random operator to exchange genes between two chromosomes and the genes might randomly be changed by mutation (Goldberg, 1989; Davis, 1991). Chromosomes are candidate solutions (selected features) which are updated in an iterative loop to find the best solution (Oh et al., 2002, 2004). GA algorithm is leveraged for feature selection in wrapper-based such that GA is coupled to SVM and the accuracy of SVM is used to evaluate the solution to find the best subset of features (Huang and Dun, 2008b). A chromosome is a solution and formed by many genes. Each gene is a binary value and only can take two values one (selected feature) and zero. The fitness function can be defined as follows:

$$f = (1 - w_F) \times (Accuracy) + w_F \times \left(\sum_{i=1}^{n_F} C_i \times F_i \right)^{-1} \tag{126}$$

where *Accuracy*, w_F, C_i and F_i are SVM classification accuracy, weight for the number of features, cost of feature i and features (It can take "1" as selected features and "0" as is not selected), respectively. C_i represents weight of feature i if prior information about feature importance is available, otherwise C_i can be excluded and same weight considered for all features. The hyperparameters of learning algorithm can be tunned simultaneously with feature selection using this technique.

SVM has hyper parameters which must be tunned. Parameter C, which is a hyperparameter to trade off between complexity of machine and the number of non-separable samples, and kernel parameter (parameter gamma) are SVM hyperparameters which can be tunned by GA simultaneously with feature selection task. Therefore, chromosomes are divided to three parts as shown in Fig. 5.

Each chromosome is binary coded, and g_C^1 - $g_C^{n_c}$, g_γ^1 - $g_\gamma^{n_\gamma}$ and g_F^1- $g_F^{n_F}$ represent the parameter "C", parameter "gamma" and selected features, respectively. n_c and n_γ are determined based on maximum value that they can take and n_F is the total number of features. The value 1 shows the selected feature and so the binary value of chromosome is working well for feature selection aim. The subtle point is that parameters "C" and "gamma" as binary value cannot be interpreted and must be transformed from genotype to phenotype.

To improve the performance of feature selection, a hybrid technique based on the mutual information maximization (MIM) and the adaptive genetic algorithm (AGA) is proposed (Lu et al., 2017). In AGA technique, crossover and mutation, as two main operations of GA, are improved based on the search space to increase the chance of obtaining the global optimal solution. Additionally, MIM multiple times are applied on data to find a subset of data and AGA is followed to obtain the optimal solution.

Since obtaining the globally optimal solution is not entirely guaranteed, a two-layer feature selection is proposed (Amini and Hu, 2021). In this technique, first layer GA is leveraged in wrapper mode and second layer ElasticNet is employed. This technique solves a feature selection problem with root mean square error (RMSE) as cost function in wrapper-phase to reduce the dimension of features and output of wrapper-phase considers as input of elastic net. As mentioned above, exploitation capability of GA is a challenge. To combat this challenge, a combined GA technique is proposed (Guha et al., 2021). Great deluge algorithm (GDA) is combined with GA to increase local search capability of GA. Mutation operation is surrogated by GDA to improve the performance of GA for feature selection problem. The idea of GDA is inspired from climbing a mountain and avoiding getting wet due to rain.

Premature convergence and exponential or high computational complexity pose as two obstacles for evolutionary algorithms. To address these challenges, a hybrid filter-wrapper evolutionary algorithm is introduced which utilizing information theory and the non-dominated sorted genetic algorithm-II (NSGA-II) (Das et al., 2022). This technique employs an initialization of NSGA-II to randomly select features, followed by the application of mutual correlation to filter out irrelevant features. The objective function of this hybrid approach shares similarities with equation (126). As previously mentioned, the utilization of a wrapper-based technique with a genetic algorithm (GA) is computationally expensive. In order to enhance the quality of the search space for NSGA-II and achieve optimal feature selection, Das et al. (2022) suggests the utilization of mutual information. Another study (Zhou and Hua, 2022) introduces a correlation-guided genetic algorithm (CGGA)

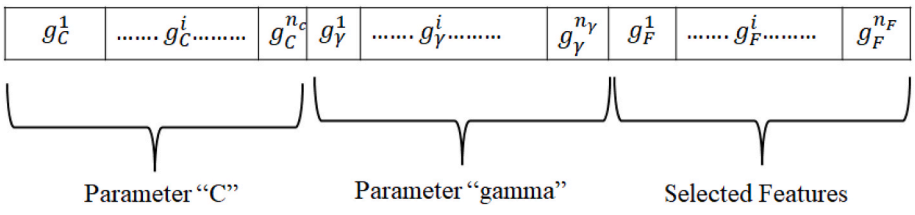


Fig. 5. A chromosome is partitioned to three parts, encompassing classifier hyperparameters and selected features. In each iteration these parts are updated.

as a means to identify the most informative features while reducing computational costs. The concept behind CGGA involves guiding the genetic algorithm using a correlation matrix. This technique involves conducting a preliminary check of potential solutions to avoid inferior ones during the crossover process. Additionally, a correlation-guided mutation operator (CGMO) is proposed and implemented to improve the quality of mutated genes. By employing CGMO, the algorithm facilitates more effective exploration of the search space.

5.2. PSO

Particle swarm optimization (PSO) is classified as a population-based, swarm intelligence-based and meta-heuristic algorithm. In PSO, population consist of particles and each particle is a subset of feature (candidate solution of cost function). Particles have two parameters: velocity and position. The position of particles carries solution of the fitness function, and the best solution is achieved in iteration mode. In each iteration, the best solution of population is considered as the best local solution and the best local is compared with global best solution. As a consequence, if local best solution is better than global, the global best solution is updated by local best solution (Kennedy et al., 1995; Clerc et al., 2002).

$$\text{MinMaxScaler} = \frac{velocity\ of\ particle - \text{minCHECKCHARACTERNOTSUPPORTED}\square(velocity)}{max\text{CHECKCHARACTERNOTSUPPORTED}\square(velocity) - \text{minCHECKCHARACTERNOTSUPPORTED}\square(velocity)} \tag{127}$$

Challenge is that PSO is a continuous algorithm and feature selection is a discrete problem. This challenge can be ameliorated by using a threshold, binary PSO (Kennedy and Eberhart, 1997) and random key trick.

In optimization problem, the combinatorial problem can be converted to continuous optimization problem by random key technique. In this technique, random numbers are generated, and these numbers are sorted in ascending order. Sorted numbers are not important, but the rank of these numbers is playing significant role to create a permutation. Random numbers are generated and sorted to create permutation and then PSO is applied to find top k features.

PSO-SVM is proposed to find the optimal subset of features and hyperparameter tuning of SVM (Lin et al., 2008). In order to determine the optimum subset of features, 0.5 considered as threshold such that feature is selected if continuous value greater than this threshold and otherwise feature is not selected. Feature selection problem can be considered as multi-objective optimization problem by considering a trade-off between classifier error and the number of selected features. Although the most of articles considered feature selection as single objective problem, nondominated sorting-based multi-objective PSO feature selection (NSPSOFS) and crowding, mutation, and dominance PSO feature selection (CMDPSOFS) algorithms are proposed as multi-objective optimization problem (Xue et al., 2013). The classifier error and number of selected features as considered as objectives. In NSPSOFS, the fitness function is computed for each particle in each iteration and nondominated solutions is recognized based on the value of cost function. All the nondominated particles positions are sorted in ascending order based on crowding distance. Global-best is updated for each solution by randomly selection from highest ranked nondominated particles. Local-best is updated by new solution if and only if the new particle dominates the current local-best solution. In CMDPSOFS, a leader set is constructed by storing the nondominated solutions as the potential leaders for each particle. Global-best is selected from leader set based on crowding distances and a binary tournament selection. In the light of feature selection by PSO, a cost-based feature selection technique based on multi-objective PSO is proposed which is called HMPSOFS (Zhang et al., 2017). In this technique, sigmoid function is used to binarize the algorithm and hybrid mutation is applied to improve the search diversity. Pareto is obtained front of non-dominated solutions, since this technique is cost-based. In the light of multi-objective PSO for feature selection, a method is proposed to rank the features and utilizing this information to refine search space with aim of algorithm improvement (Amoozegar and Minaei-Bidgoli, 2018a). The fitness function is defined as combination of classifier accuracy and feature ratio (the number of selected features normalized to the total number of features). An external archive is defined to store non-dominated solutions such that in each iteration nondominated solutions is obtained based on fitness function. Archive set is updated based on new non-dominated solutions. In next step, the features are ranked by archive members and archive set is refined by adding high rank features to generate new population for updating the local-best and global best. In the last step, mutation mechanism is leveraged to increase diversity in archive set.

Trapping in local minima is found as main drawback of PSO which directly impacts on classifier performance by selecting non-optimal features. To remedy this problem, MinMaxScaler binary is proposed (Shaheen et al., 2020). MinMaxScaler is a scalar and defined as follows:

The position of particle is binarized and gets value “1” (selected feature) if the MinMaxScaler is greater than 0.5, otherwise gets value “0”. The performance of feature selection can be improved by merging PSO feature selection with other feature selection techniques. To this end, a three-phase online feature selection method is proposed (Paul et al., 2021).

First phase, multi-objective PSO feature selection, which is called intra-batch filtering, is leveraged to select the relevant subset of features withing groups.

Second phase, inter-batch filtering is utilized to remove the redundant features.

Third phase, non-significant features are excluded.

Minimum redundancy and maximum relevancy are embedded in this multi-objective feature selection technique to improve the performance. The normalized redundancy of selected features (Red_S) is defined as follows:

$$Red_S = \sum_{f_i \epsilon S} \sum_{f_j \in S, f_i \neq f_j} \frac{NMI(f_i, f_j)}{n_{sel} \times (n_{sel} - 1)} \tag{128}$$

where S and n_{sel} are the selected features set and the number of selected features, respectively. Normalized mutual information is defined as follows:

$$NMI(f_i, f_j) = \frac{MI(f_i, f_j)}{\sqrt{E(f_i) \times E(f_j)}} \tag{129}$$

where E is entropy and f indicates the feature. The normalized redundancy of non-selected features (Red_{NS}) is defined as follows:

$$Red_{NS} = \sum_{f_i \epsilon NS} \sum_{f_j \in S,} \frac{NMI(f_i, f_j)}{n_{sel} \times n_{notsel}} \tag{130}$$

where NS and n_{notsel} are non-selected features set and the number of non-selected features, respectively. Although Red_S and Red_{NS} provide the

redundancy information about features, the relevancy information is playing significant role to find more informative features. To this end, the relevancy of selected features with labels consideration is defined as follows:

$$Rel_S = \frac{\sum_{f_i \epsilon S} \sum_{l \epsilon L} NMI(f_i, l)}{L \times n_{sel}} \tag{131}$$

where L is the number of labels. Eventually, Multi-objective PSO as first phase of this technique can be expressed via (129) and (130).

$$Red_S - Red_{NS} \tag{132}$$

$$Red_S \,/\, Rel_S \tag{133}$$

This can be implied that (128) and (129) are aimed to minimize the redundancy in selected features and maximize the relevancy between features and labels. The redundancy is maximized in the set of non-selected features in (131-133). Second phase, it is possible to find redundancy between two features in two different feature sets (groups). As a result, in this phase two thresholds are defined such that first threshold is maximum normalized mutual information (MNMI) between two features among all selected features from phase one, and second threshold is MNMI between a selected feature and rest of the features. Consequently, the feature is selected in second phase if the second threshold is greater than the first threshold, otherwise feature will be discarded. The last phase of this multi-phase feature selection is non-significant feature filtering. In this phase, rough set theory is leveraged to measure the vagueness of boundary region of selected features set.

As aforementioned, the feature selection problem is a discrete problem, while PSO is a continuous algorithm. To this end, PSO should be binarized. This point should be noted that the performance of algorithm can be affected by the cut-point threshold for binarization. Each particle in PSO has value in range [0,1], and cut-point can be any value in this range. As result, a good cut-point is a big challenge, and it needs to be tuned during feature selection. In this regards, potential PSO (PPSO) is proposed to discretize PSO and select top features in a single stage (Tran et al., 2018). PPSO is inspired from bare bones PSO (BBPSO) that is operating PSO by sampling in search space. PPSO provides a potential cut-point set for each feature using entropy-based such that minimum description length principle (MDLP) must be satisfied. This is important to note that the number of potential cut-point for each feature can be different. Particle position in PPSO is an integer value and the cut-point index indicates the position within the potential set of cut-points where a cut-point is selected. If the index is out of range, the corresponding feature should be overridden. The fitness function has two parts including 1) classification accuracy and 2) distance value among the features.

Evolve PSO (EPSO) is another technique based on BBPSO that is proposed for feature selection (Tran et al., 2016). In EPSO, cut-point is directly evolved and the range is defined form minimum value of features to maximum value of features. The subtle point of EPSO is that if cut-point is placed out of range, the nearest boundary will be considered as cut-point.

The hybrid technique is another approach to improve the performance of PSO for feature selection. To this end, hybridization GA and PSO (HGAPSO) is proposed to increase the search capability and decrease the computation time (Pedram Ghamisiand Jon Atli Benediktsson, 2015). In this technique, SVM is coupled to HGAPSO such that the accuracy of validation set is used as fitness function. The particles (selected features) are ranked based on accuracy of SVM and three actions including enhancement, crossover and mutation, are applied to produce the next generation of solutions (particles). In enhancement step, the top-half particles are considered as elites and are enhanced by PSO. The enhanced solutions are considered as parents to generate offspring. Next step, the crossover and mutation are applied on selected features by PSO. Eventually, the output of PSO and GA are merged to generate updated solutions. Fig. 6 shows the flow of HGAPSO.

In the light of PSO-GA hybrid technique, a method to use PSO in parallel with GA is proposed (Moslehi and Haeri, 2020). In this algorithm, the features are ranked based on gain ratio and 85% of features with highest gain ration are selected. In next step, the solutions for PSO and GA are randomly generated, and error of artificial neural network (ANN) is used to evaluate the solution. The solutions of PSO, GA, crossover of GA and mutation of GA are merged to provide a set of solutions. The best solutions are extracted from integrated solutions.

Getting stuck in local minimum is considered as main drawback of PSO. To ameliorate this challenge, micro-GA embedded PSO is proposed to increase performance of PSO for feature selection task (Mistry et al., 2017). This technique alleviates the premature convergence challenge of PSO. As a consequence, the most important features which carry the discriminative information can be selected. In micro-GA embedded PSO, averaging search and gaussian mutation are leveraged for local-best and global best updating, respectively. A copy of initialized primary swarm is stored in a non-separable memory to increase swarm diversity during algorithm updating. A secondary swarm with five particles, one leader and four followers, is generated from non-separable memory in order to increase capability of searching for both local and global. Leader of particles is best global, and four followers are selected from non-separable memory such that they have highest correlation with leader.

5.3. ACO

Ant colony optimization (ACO) is another metaheuristic algorithm that is inspired by behavior of ants to find shortest path of food resources. Feature selection problem in order to solve by ACO can be represented as graph such that nodes are representing features and edges

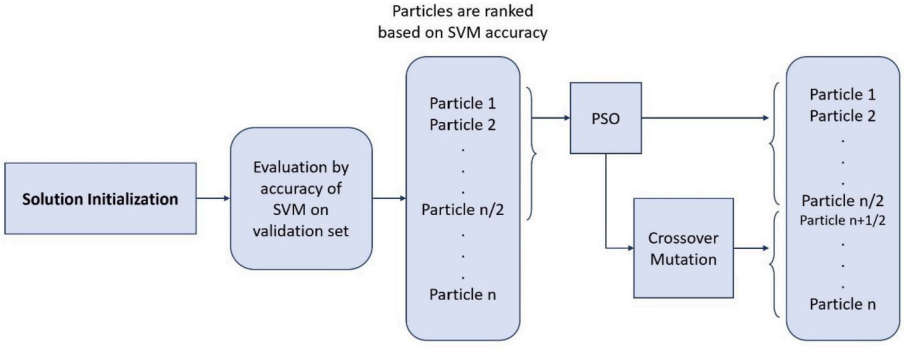


Fig. 6. Schematic of HGAPSO for feature selection. In this technique, particles of lower significance undergo crossover and mutation to enhance exploration and exploitation. Top half particles are directly considered for next iteration and rest half are considered for next iteration after undergoing crossover and mutation.

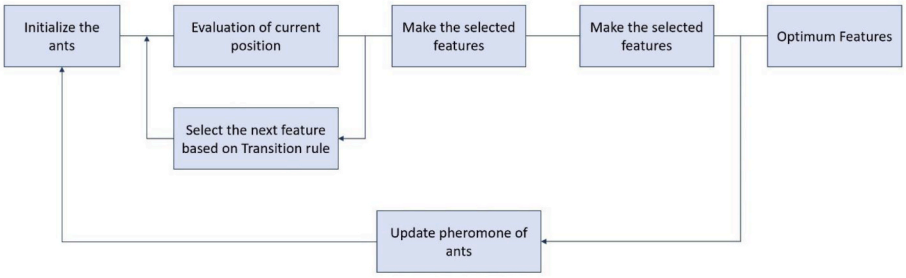


Fig. 7. Schematic of ACO feature selection. This figure shows the different steps to select the best feature by ants.

between nodes are choice of the next selected feature. The next node (feature) is selected by ant based on the transition rule. Entropy-based measure, rough set dependency measure, the fisher discrimination rate and probabilistic-based measure are transition rules that ant can make decision to choose the next node and add to its path (Kanan and Faez, 2008). Fig. 7 illustrates the schematic of ACO feature selection.

Search space for high dimensional data is extremely expanded and then searching time for ants to find optimum path is subsequently increased. In this situation, finding the optimal features subset can be difficult task due to the number of features. To combat this challenge, two-stage hybrid ACO for high-dimensional feature selection (TSHFS-ACO) is proposed (Ma et al., 2021). The TSHFS-ACO embraces two stages and leverages advanced hybrid ACO algorithm. In this technique, the correlation between features is measured to remove redundant features and symmetrical uncertainty (SU) is applied for pheromone initialization (initial exploration direction) to avoid a large number of dispensable features consideration. A greedy-probabilistic transition rule is used to balance the local search and the global search, and max-min ant system (MMAS) is applied for pheromone updating. The combination of accuracy of KNN as classifier and distance among features are considered as fitness function. Likewise, distance function is applied to minimize distance for same class of samples and maximize distance for different class of samples. This feature selection technique follows two steps: first step is to determine the optimal number of features and second step is feature reduction. In step one, three intervals are defined for features including correlation interval, redundant interval and uncorrelated interval. The optimal number determination in each interval has discontinuity that make problem to have perfect search. To ameliorate discontinuity problem, gradient descent idea is added to interval method. In each interval the end points are evaluated to reduce computation time. Second step, feature reduction is applied on optimal featurs set that is constructed in step one. A probability of removing for each feature is calculated to refine features set and find the optimal features to obtain the maximum accuracy of classifier.

5.4. GWO

Grey wolf optimization (GWO) as one of the new continuous bio-inspired optimization techniques that is inspired from strategy of grey wolves to hunt the prey. GWO must be binarized for feature selection since the feature selection is a discrete problem. To this end, positions of wolves are binarized by using sigmoid function. Therefore, binary GWO (BGWO) can be constructed with aim of feature selection (Emary et al., 2016). BGWO can be applied for feature selection by solving following fitness function;

$$\gamma(Accuracy) + (1-\gamma)\frac{\left|length_{Total\ features} - length_{Selected\ features}\right|}{\left|length_{Total\ features}\right|} \tag{134}$$

where *Accuracy* shows the classification quality, $0 < \gamma < 1$ is a hyper-parameter to keep a trad-off between accuracy of classification and the number of selected features. The subtle point should be noted is that $\frac{\left|length_{Total\ features} - length_{Selected\ features}\right|}{\left|length_{Total\ features}\right|}$ represents the ratio of unselected features and then fitness function (134) is categorized as maximization problem and $|.|$ is cardinality of set.

Binarization of GWO is important to improve the performance of algorithm. Transfer function for converting continuous values to discrete is playing significant role in feature selection problem. To this end, different V-type transfer functions for feature selection are applied (Hu et al., 2020). $\frac{|\tanh(x)|}{\tanh(4)}$, $\left|\frac{\sqrt{17}\times x}{4\times\sqrt{1+x^2}}\right|$ and $\left|\frac{\arctan\left(\frac{2}{\pi}x\right)}{\arctan(2\pi)}\right|$ are proposed as improved transfer functions. Therefore, following cost function is applied for feature selection;

$$\gamma(Error\ of\ classification) + (1-\gamma)\frac{|Selected\ features|}{|Total\ features|} \tag{135}$$

where (135) is minimization problem and this study considered $\gamma = 0.99$. Additionally, $|Selected\ feature|$ and $|Total\ feature|$ are the number of selected features and total features, respectively.

6. Reinforcement learning for feature selection

Reinforcement learning (RL) has been a widely discussed and popular topic in the field of machine learning, and it can be applied in game theory, statistics, combinatorial optimization problems, robotic and multi-agent state. Additionally, it can be leveraged for feature selection. Regarding the components of reinforcement learning (RL), the RL framework involves an agent that learns directly from the environment through a trial and error mechanism. At each step, the agent selects an action to interact with the environment, causing the state of the environment to change accordingly. Subsequently, the agent receives feedback from the environment in the form of a reward, which evaluates the outcome of each transition. Final goal of agent is to maximize the rewards (Kaelbling et al., 1996). Fig. 8 illustrates the mechanism of RL.

RL problem can be modeled as a Markov decision process (MDP) and it includes four main parts: 1) a finite state-space, 2) S that determines the set of actions which can be taken from each state by agent, 3)

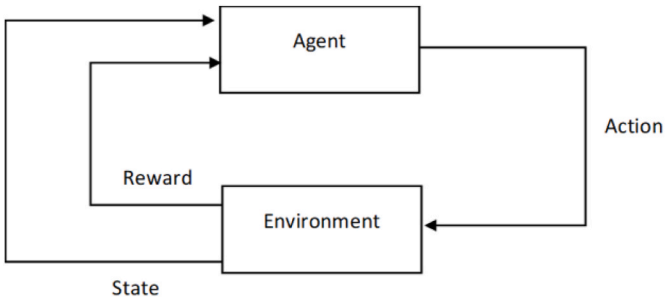


Fig. 8. Schematic of interaction between agent and environment. Agent determines its next action based on its previous experience and associated reward. Agent learns the optimum actions in direct interaction with environment.

transition function T, and 4) reward function R which shows the amount of reward for agent based on taken action (Sutton and Barto, 2018). MDP problems are solved by dynamic programming (DP), Monte Carlo (MC) methods, and temporal difference (TD) learning approaches. TD approaches leverage step-by-step increasing computation to calculate value of current state and next step. Consequently, TD can be appropriate approach for feature selection task. In RL-based feature selection, the agent starts to select feature in each iteration based on predefined distribution and its prior experience. Then, learning algorithm is trained by selected features and the performance of machine is considered as memory for agent to learn about space. Therefore, feature selection task can be modeled as a MDP and features can be sequentially selected until termination state meet by agent based on greedy strategy.

Concisely, all features are considered as state space and action (selecting the subset of unseen features) is considered as transition from one state to another. As a result, in each iteration unseen features are added to an empty set to find most important features based on specified policy (Gaudel and Sebag, 2010).

RL problem has four components including: policy (π), reward (r), value function (V) and the environment model. The behavior of agent in the environment and specific time is determined by policy. Quality of agent's action is indicated by reward signal. The main goal is to maximize the total reward.

To this end, policy must be adapted based on reward for future action. The cumulative rewards obtained through the agent's actions are regarded as the value of a state, while the value function assesses the quality of states based on how advantageous they are for the agent.

In the context of feature selection using RL, ACO and temporal difference RL combined to solve the feature selection problem (Paniri et al., 2021). To this end, feature selection problem is modeled as a MDP problem such that features represent the states, and the subset of features, which is selected by an ant, represents the action. Since the aim of feature selection is to minimize the redundancy among the features and maximize the relevancy between features and labels, correlation among features (redundancy) and cosine similarity between features and labels (relevancy) are considered as function to measure reward signals for ant's action (selecting feature). Since ACO algorithm usually follows fixed heuristic pattern and it does not change during search process, TD (0) RL is leveraged to learn optimal value function (V) as heuristic information. It means that heuristic can be used from the experience of all ants in each iteration to improve the performance. TD (0) value function is defined as follows:

$$V(s_t) = V(s_t) + \alpha[r_{t+1} + \gamma V(s_{t+1}) - V(s_t)] \tag{136}$$

where α and γ are the learning rate, which determines the amount of accepted error in each step, and discount rate which is used to specify the effect of next step, respectively (α; $\gamma \in [0, 1]$). Reward function is defined by combining the cosine similarity (CosSim) function and correlation (Cor) function. Consequently, reward function can be expressed as follows when agent moves form state t to state $t+1$ by taking action a:

$$r_{t+1} = \frac{\max_{s_{t+1}}\left(CosSim_{s_{t+1},l}\right)}{1 + Cor_{s_t,s_{t+1}}} \tag{137}$$

where $CosSim_{s_{t+1},l}$ and $Cor_{s_t,s_{t+1}}$ are cosine similarity between next feature and all of its corresponding labels, and correlation between next feature and current feature, respectively. In each iteration, ants construct their solutions by selecting features and update function V locally. As a result, V globally updated at end of iteration by getting average on vector-V of all ants. In terms of solutions construction, ants select features randomly, and each ant constructs its solution using different features in search space by applying greedy and probabilistic transition rules.

A vector which is called ant feature count (AFC) is defined to construct the path of ants by visiting features. Therefore, AFC vector will be incremented by visiting a feature. Then, pheromone is updated for each feature i as follows:

$$pheromone_i(t+1) = (1-\rho)pheromone_i(t) + \frac{AFC(i)}{\sum_{j=1}^{d} AFC(j)} \tag{138}$$

For initialization, maximum mutual information value between features and labels and maximum cosine similarity between each feature and all labels are considered to initialize the pheromone and state-value vector V, respectively.

Eventually, features are ranked based on pheromone values in descending order and top k-feature can be selected.

In another study, feature selection based on Q-learning is proposed (Xu et al., 2021). In this technique, all the state-action combinations are stored in Q-table, and the function approximation method is leveraged to approximate Q-table. In this method, Q function is specified by state (s), action (a) and discriminator vector. A discriminant function vector is constructed by Q-learning such that each discriminator vector w in discriminant matrix W is updated based on corresponding selected action and environment feedback. In this technique, actions and states are separately encoded such that Q-function is calculated based on an action which is selected to predict specified label. Greed strategy is adopted to select actions and Q-function is calculated using polynomial expression as follows:

$$Q(s, a_i) = w_i \times \varphi(X) \tag{139}$$

where $\varphi(x)$ is the expression of specified state corresponding to sample X and a is obtained as follows:

$$a = \begin{cases} argmax_a\ Q(s,a) & 1-\varepsilon \\ select\ actions\ randomly & \varepsilon \end{cases} \tag{139a}$$

For clarification, a sample with label i, value of $w_i \times \varphi(X)$ is positive and $w_j \times \varphi(X)$ is negative because it has different label. Therefore, action a_i is selected for a sample and label is checked, if it has same label, it will obtain reward and w_i will be updated, otherwise will obtain punishment.

Finally, $W \epsilon R^{m \times c}$ (m and c are the number of features and number of labels, respectively.) as discriminant function is considered to rank the features such that Euclidean norm of each row is calculated for matrix W.

In another study, Karimi et al. (2023) proposed a semi-supervised feature selection based on ant colony and reinforcement learning. This technique combined TD reinforcement learning and ACO to select the most informative features by considering the minimum redundancy among the features and maximum relevancy between features and labels. Zhang et al. (2023) proposed a RL-based feature selection technique which is called Q-learning based on collaborative agents (QLCA). This technique utilizes two collaborative agents including selection agent and classification agent to obtain the most discriminative features. Selection agent considers Fisher score as reward to select feature and classification agent considers the error of classifier as feedback to obtain the optimum number of features.

Table 5
Details of the nine gene expression common datasets.

Dataset	# of samples	# of features	# of classes
CNS	60	7129	2
Colon	62	2000	2
DLBCL	47	4026	2
GLIOMA	50	4434	4
Lymphoma	96	4026	9
TOX-171	171	5748	4
Leukemia	72	7070	2
Prostate-GE	102	5966	2
CCLE	343	1899	2

7. Common datasets

This section presents a selection of commonly used benchmark datasets that can be employed for evaluating feature selection methods. There is a line of research to select genes in DNA microarray datasets. These datasets are shown in Table 5.

The CNS (Embryonal tumors of CNS) dataset (Pomeroy et al., 2002) comprises gene expression data obtained from 60 patients with various types of brain tumors. The Colon dataset (Alon et al., 1999) consists of gene expression signatures from 40 tumor cells and 22 normal colon cells. The DLBCL (Diffuse Large B-cell Lymphomas) dataset (Alizadeh et al., 2000) contains information about tumor cells collected from patients with immune system cancer. The GLIOMA dataset (Li et al., 2018) includes 50 samples categorized into four classes: cancer oligodendrogliomas, non-cancer oligodendrogliomas, cancer glioblastomas, and non-cancer glioblastomas. The Lymphoma dataset (Li et al., 2018) focuses on gene expression related to the lymphatic system and encompasses various forms of non-Hodgkin's lymphoma. The TOX-171 dataset (Stienstra et al., 2010) consists of 171 samples, each with 5748 genes, and includes classes such as patients with no cancer, controlled radiation-therapy patients, patients with skin cancer, and patients who have undergone radiation therapy. The Leukemia dataset (Golub et al., 1999) originates from a study on a group of leukemia patients and contains 72 samples with 7070 genes. Prostate-GE (Li et al., 2018) is another gene expression dataset comprising 102 samples and 5966 genes. Finally, the Cancer Cell Line Encyclopedia (CCLE) (Barretina et al., 2012) is a well-known dataset extensively used in cancer-related biomedical research, focusing on standard cancer cell lines.

There are 14 various types of high-dimensional real-world datasets including: 20News: the 20 Newsgroups datasets which are collected from approximately 20 000 newsgroup documents,[1] WebKB: the world wide knowledge base dataset4,[2] SRBCTML: the small round blue cell tumors dataset,[3] PubFig: the public figure face image set (Xu et al., 2018), OSR: the outdoor scene recognition dataset (Parikh and Grauman, 2011), Wiki: the Wikipedia dataset (Rasiwasia et al., 2010) is based on Wikipedia's featured articles, orlraws10P and warpPIE10P,[4] one object image dataset COIL20,[5] Yale and spoken letter Isolet[6] and Palm25.[7] Table 6 shows these datasets with description.

Table 6
Real world datasets description.

Datasets	# of features	# of samples	# of classes	Type
20News	8014	3970	4	Text
orlraws10P	10304	100	10	Face Image
WebKB	4165	1166	7	Web
warpPIE10P	2420	210	10	Face Image
Mnist	784	6996	10	Digits
SRBCTML	2308	83	4	Biology
COIL20	1024	1440	20	Object Image
Pubfig	65536	772	8	Face image
OSR	65536	2688	8	Scene image
Wiki	512	2866	10	Image & Text
Yale	1024	165	15	Image
Isolet	617	1669	26	Voice
Palm 25	256	2000	50	Image

[1] http://qwone.com/~jason/20Newsgroups/.
[2] http://www.cs.cmu.edu/afs/cs/project/theo-20/www/data/.
[3] http://www.biolab.si/supp/bi-cancer/projections/info/SRBCT.html.
[4] http://featureselection.asu.edu/datasets.php.
[5] http://www.cs.columbia.edu/CAVE/software/softlib/coil-20.php.
[6] http://featureselection.asu.edu/datasets.php.
[7] http://www.escience.cn/people/fpnie/index.html.

8. Discussion

There has been a considerable focus on feature selection to enhance data interpretability and reduce complexity. Various techniques for feature selection can be chosen based on the data type, availability of labels, and data dimensionality. Feature selection covers a broad range of topics, such as subspace learning, information theory, ensemble learning, and evolutionary computation.

When it comes to subspace learning, there are two primary approaches: matrix factorization and projection to a subspace using an orthonormal projection matrix. Feature selection based on matrix factorization aims to identify a high-quality subspace within a data matrix that spans the original data. By formulating a minimization problem that seeks to minimize the reconstruction error between the data matrix and the subspace, it becomes possible to extract low-dimensional data with maximum information. Nonnegative matrix factorization-based techniques are frequently utilized as unsupervised filter-based techniques. In matrix projection as another approach of subspace learning, a projection matrix with orthogonality constraint is considered to obtain subspace of original data. Latent representation learning and regression-based models are frequently following matrix projection techniques.

To improve the quality of extracted subspace, two types of regularization terms, namely global structure learning and local structure learning, can be incorporated into the cost function. This addition aims to improve the performance of the feature selection algorithm. It is essential to test the convergence of algorithm and compute complexity of algorithm to ensure proper application and computation, respectively. Since these algorithms have an iterative body, complexity analysis is essential which is directly associated with computation time. Most of the articles have focused on utilizing Laplacian graphs to retain the geometric details of samples. However, it is important to note that using Laplacian graphs can introduce a bias towards a constant geodesic function and cannot perfectly preserve the topological structure (Shi et al., 2015; Kim et al., 2010). In this regard, the proposal suggests employing Hessian regularization as a structure learning regularization method, as it exhibits superior extrapolation capabilities when compared to Laplacian graphs. The null space associated with Hessian regularization is more diverse than that of Laplacian graphs, making it suitable for functions that exhibit linear changes based on geodesic distance (Shi et al., 2015; Kim et al., 2010). Consequently, Hessian regularization, with its richer null space, can more effectively preserve the topology of the data. The trace of hessian regularization can be found colorfully in semi-supervised feature selection (Sheikhpour et al., 2023).

The convexity of the objective function is a crucial consideration that should be taken into account. In the case of $l_{2,1-2}$, it is not convex, necessitating the use of the CCCP (Concave-Convex Procedure) to linearize the concave part and achieve a convex objective function. However, it is important to note that linearizing in each iteration results in increased computation time. Similarly, the inner product norm $(\|WW^T\|_1 - \|W\|_2^2)$ also poses challenges in terms of nonconvexity. However, many techniques that utilize this norm have not effectively addressed the issue of nonconvexity.

Feature selection can be achieved by employing sparse representation learning techniques. By utilizing compressed sensing theory and dictionary learning, which are part of sparse representation learning, it is possible to identify the most informative feature. This is accomplished by constructing a sensing matrix and a dictionary matrix. In the field of engineering, sensor localization poses a combinatorial and NP-hard problem. However, sparse representation learning has been applied to determine the optimal sensor locations. These techniques can be adapted to the domain of feature selection, where sensors are represented by features. Brunton et al. (2016) introduced a sparse representation learning approach based on compressed sensing to identify the optimal sensors. They treated each pixel of an image as a sensor, thereby considering each pixel as a feature.

Evolutionary computation techniques are widely applied for feature selection problems. However, these techniques often encounter the problem of getting stuck in local optima. Additionally, their computational costs are high, primarily due to their predominant reliance on the wrapper strategy. Assigning an appropriate fitness function is the most important part of feature selection using evolutionary computation techniques. These algorithms are able to be applied as multi objective optimization to improve the classifier performance, decrease the computational cost, and adjust the curse of dimensionality simultaneously (Amoozegar and Minaei-Bidgoli, 2018b). These techniques are able to tune the hyperparameters of machine learning algorithms which is the prominent advantage of these techniques. Nevertheless, getting stuck in local minimum is still a remaining challenge for these methods. Multicollinear features often play role as local minima, posing challenges for evolutionary computation techniques. Hybrid feature selection can be an effective approach to tackle this challenge by filtering out redundant features. In hybrid techniques, redundant features can be removed by other techniques such as NMFFS and then PSO, GA, ACO and GWO can be applied to find the optimum subset of features. For instance, Moslemi et al. (2022b) combined RRE and GA to construct a hybrid feature selection such that redundant features are removed by RRE and then GA is applied to find optimum features. Moreover, evolutionary computation techniques possess the unique capability of determining the optimal number of selected features. Generally, the performance of metaheuristics algorithms for feature selection can be significantly improved by providing a good balance between exploitation and exploring. For RL-based techniques, the performance is directly affected by policy.

In comparison with other feature selection surveys, Gui et al. conducted a comprehensive survey on Feature Selection based on structure sparsity (Gui et al., 2017). This survey only considered Sparsity structure learning regularization. Whereas several studies showed that both structure sparsity and local structure learning should be applied to keep maximum information after projecting to subspace. The lack of local structure learning results in the loss of geometric information from the samples. Additionally, in our survey, new matrix norms $\|\boldsymbol{W}\|_{2,1-2}$, $\|\boldsymbol{W}\|_{2,1/2}$, $\|\boldsymbol{W}\|_{2,0}$ and inner-product norms were discussed, and advantages and weakness of each norm comprehensively were discussed. Solorio-Fernandez et al. conducted a survey on only unsupervised feature selection (Solorio-Fernandez et al., 2020), but we discussed supervised, unsupervised, and semi-supervised feature selection. Furthermore, we derived mathematical formulas and showed algorithms of fundamental techniques which distinguish our survey from Fernandez survey. Zhang et al. conducted a survey on feature selection for multi-view data (Zhang et al., 2019a). Although there is similarity between this survey and our survey, they only focused on the application of feature selection in information fusion domain and did not cover evolutionary algorithms and reinforcement learning. Xue et al. conducted a comprehensive survey on genetic algorithm and PSO, as evolutionary computation techniques for feature selection (Xue et al., 2016). This survey analyzed weakness and strength of evolutionary techniques for feature selection comprehensively, but we analyzed the GWO and ACO in addition to Genetic algorithm and PSO. Sheikhpour et al. (2017) conducted a survey on semi-supervised feature selection, which involved an examination of various categories of feature selection methods, including filter, wrapper, and embedding techniques. However, our survey goes beyond that by encompassing not only all types of feature selection techniques based on label information but also incorporating different approaches such as evolutionary computation and reinforcement learning. Furthermore, the application of compressed sensing is well-known for its ability to reconstruct signals using limited measurements. In this survey, we included articles that have explored the application of compressed sensing for feature selection. Dokeroglu et al. (2022) conducted a comprehensive survey on metaheuristics-based feature selection techniques. In this survey, several metaheuristics optimization techniques which can be applied for feature selection tasks have been encompassed. These algorithms are Harmony Search, Bacterial Foraging Optimization, Artificial Fish Swarm Algorithm, Artificial Bee Colony Algorithm, Biogeography-Based Optimization, Firefly Algorithm, Gravitational Search Algorithm, Cuckoo Search, Bat Algorithm, Teaching–Learning-Based Optimization, Krill Herd, Social Spider Optimization, Grey Wolf Optimization, Ant Lion Optimization, Sine Cosine Algorithm, Whale Optimization Algorithm, Crow Search Algorithm, Dragonfly Algorithm, Salp Swarm Algorithm, Grasshopper Optimization Algorithm, Butterfly Optimization Algorithm and Harris' Hawk Optimization. This survey is a comprehensive review of feature selection techniques using metaheuristics algorithms. However, our survey considered a wide spectrum of feature selection techniques including matrix-based techniques, sparse representation learning techniques, information theory-based techniques, metaheuristics-based techniques and reinforcement learning-based techniques. Additionally, combination of reinforcement learning with metaheuristics techniques for feature selection is a recent topic which is received attention.

Our survey takes the form of a tutorial, focusing on providing a comprehensive overview rather than comparing different methods. The primary objective is to demonstrate the process of constructing and implementing a feature selection algorithm, as well as exploring avenues for improvement.

9. Conclusion

This paper provided a comprehensive tutorial survey of feature selection. This study encompassed feature selection tasks using techniques such as matrix factorization-based methods, sparse representation learning, information-based approaches, evolutionary computation-based techniques, and reinforcement learning (RL)-based methods. As a result, this study summarized and elaborated the state-of-the-art techniques of feature selection. This survey not only introduces various feature selection techniques but also provides mathematical expressions and algorithms for each technique, along with discussing their limitations. Based on the literature review, Matrix-based techniques can be utilized for supervised, semi-supervised, and unsupervised problems. In contrast, a majority of information theory-based techniques, evolutionary techniques, and reinforcement learning-based techniques are predominantly supervised and require labeled information. Particularly, in RL and evolutionary techniques, the classifier error is employed as feedback or a reward signal to update the algorithm.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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